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Research Project Report on the Topic:
Optimisation of Warehouse Slotting and Routing

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Abstract

This paper studies warehouse order picking optimisation through the integration of analytical inventory classification and spatial movement analysis. Traditional approaches rely either on ABC×XYZ analysis or on spatial tools such as RTLS tracking and heat maps, but these methods are usually applied independently, which limits their effectiveness.

The study proposes a unified framework that combines ABC×XYZ analysis, simulated RTLS data, heat-map-based congestion modelling, and routing algorithms in a single graph-based model, where edge weights include both physical distance and congestion penalties.

Computational experiments on synthetic warehouse data show that the proposed method reduces both average route length and aisle congestion compared with non-integrated approaches in the tested single-depot setting.

В работе изучается задача оптимизации комплектации заказов на складе через интеграцию аналитической классификации запасов и пространственного анализа перемещений. Традиционные подходы опираются либо на ABC×XYZ-анализ, либо на пространственные инструменты мониторинга (RTLS, тепловые карты), но обычно применяются раздельно, что снижает их эффективность.

В работе предлагается единая модель, объединяющая ABC×XYZ-анализ, симулированные данные RTLS, построение тепловых карт и алгоритмы маршрутизации в одном графовом представлении, где вес ребра зависит как от расстояния, так и от загруженности.

Результаты вычислительных экспериментов на синтетических данных показывают, что предложенный метод снижает как средний путь сборки, так и уровень загруженности проходов по сравнению с неинтегрированными подходами в исследованной конфигурации склада с одним пунктом приёма/выдачи.

Keywords

Warehouse optimisation; ABC×XYZ analysis; slotting problem; congestion-aware routing; RTLS simulation; heat map analysis; order picking; graph-based modelling.

1 Introduction

Warehouse operations constitute a critical component of modern supply chains. Storage organization, order picking, and internal transportation directly influence operational costs, service levels, and overall logistics performance. In many distribution centers, order picking accounts for a significant share of total warehouse operating costs, making it one of the primary targets for optimisation.

Traditionally, warehouse management relies on analytical inventory classification methods such as ABC and XYZ analysis. ABC analysis ranks items according to their economic importance, typically based on turnover or contribution to revenue, while XYZ analysis groups items by demand stability. The combined ABC $\times$ XYZ framework enables structured prioritisation of stock keeping units (SKUs) and supports inventory planning decisions. However, these methods do not consider the spatial layout of the warehouse or the dynamic movement of operators and equipment.

Supply chain disruptions — most visibly those triggered by the Covid-19 pandemic — have shifted the strategic focus of logistics from pure cost minimisation toward operational resilience and flexibility, accelerating the adoption of data-driven decision support across all supply chain tiers [5]. At the warehouse level, this pressure manifests as a growing need to respond to volatile demand patterns while maintaining service levels, making the integration of analytical and spatial tools more urgent than ever.

At the same time, advances in digital technologies have enabled the collection of detailed spatial-temporal data through real-time location systems (RTLS) and indoor positioning systems (IPS). These technologies allow warehouses to monitor movement patterns, identify congestion zones, and construct activity heat maps. Spatial analysis provides insight into operational bottlenecks and traffic intensity within aisles. Nevertheless, such tools are often applied separately from analytical inventory models and mainly serve monitoring rather than integrated optimisation purposes.

The separation between demand-based prioritisation and spatial congestion analysis limits the effectiveness of warehouse optimisation strategies. Classical ABC $\times$ XYZ slotting improves placement decisions but ignores aisle load and traffic interactions. Conversely, congestion-aware routing can reduce local bottlenecks but does not differentiate between high- and low-priority SKUs. As a result, warehouse systems frequently optimize isolated components rather than the entire process.

This study focuses on small-to-medium warehouses (approximately 1 000–5 000 m², 500–5 000 SKUs) that are common among regional third-party logistics (3PL) providers, e-commerce fulfillment centres, wholesale distributors, and spare-parts depots. A key characteristic of this segment is the single-depot configuration: receiving and dispatch share the same access point, and pickers follow a return-to-origin route. This configuration corresponds to the classical return routing problem studied since Ratliff and Rosenthal [11] and further developed by Petersen [12], as opposed to the dual-depot setups typical of large-scale distribution centres with separate inbound and outbound docks.

Within this operational setting, the separation between demand-based and spatial optimisation

methods is particularly problematic. A single chokepoint near the depot concentrates nearly all picker traffic, so even a well-classified inventory will generate heavy congestion if spatial constraints are ignored. The present study addresses this limitation by proposing an integrated framework that simultaneously accounts for inventory importance, demand variability, and spatial congestion. The warehouse is modelled as a graph, where nodes represent pick locations and intersections, and edges represent feasible movements. Heat map data derived from simulated RTLS tracks are incorporated into edge weights as congestion penalties. Slotting decisions are informed by ABC×XYZ classification, while routing is performed using congestion-aware shortest-path algorithms.

The innovative contribution of this work lies in the explicit integration of analytical inventory prioritisation with spatial-temporal congestion modelling within a single optimisation framework. Unlike existing approaches that treat these components independently, the proposed model evaluates their combined impact on operational performance.

Structure of the paper. The remainder of the report is organised as follows. Section 2 reviews the relevant literature on inventory classification, RTLS-based spatial tracking, slotting strategies, and order picking routing, and positions the present work in this context. Section 3 gives the conceptual overview of the proposed method. Section 4 provides the formal problem formulation (sets, decision variables, cost structure, and objective function). Section 5 describes the experimental setup: the warehouse layout, the SKU dataset, the ABC×XYZ classification, the baseline graph model, and the 2×2 experimental design with its four evaluation metrics. Section 6 reports the results for Config A (congestion-aware routing on classical slotting), including a sensitivity analysis on the congestion penalty λ . Section 7 reports the results for Config B (heat-map-informed slotting under shortest-path routing) and compares six concrete slotting strategies (DWC, HZR, TAPS, Aisle-Balanced, AAHMS, and the exploratory CAS). Section 8 reports Config C, which closes the 2×2 design by combining the two interventions, and reports the full 7×2 matrix of slottings under both routing algorithms. Section 9 concludes with the main findings and limitations, and Section 10 outlines directions for future work.

2 Overview of Bibliographical Sources

The reviewed literature covers four major research directions relevant to warehouse optimisation: inventory classification methods, RTLS-based spatial tracking, slotting optimisation, and warehouse digitalisation.

Trubchenko et al. [1] apply ABC×XYZ analysis to the inventory of Karandash LLC, a Russian commercial enterprise with approximately 30 000 office supply SKUs. ABC analysis ranks items by annual sales volume, while XYZ quantifies demand stability using the coefficient of variation $V = 100\sigma/\bar{q}$, with thresholds $X \leq 10\text{--}15\%$, $Y = 15\text{--}25\%$, and $Z > 25\%$. The study finds that all analysed items fall into Group Z, yielding three cross-groups — AZ, BZ, and CZ — each with a dedicated replenishment strategy. Applying these recommendations reduced inventory

storage costs by 20%. Despite the practical impact, the method remains purely analytical: it does not incorporate the spatial layout of the warehouse, movement patterns of operators, or operational congestion effects, limiting its applicability to static inventory planning scenarios.

Pekarčíková et al. [2] investigate simulation-based optimisation of material supply using RTLS data in automotive production environments. Their work integrates real-time tracking data with simulation models in order to analyze spatial-temporal movement patterns, identify bottlenecks, and evaluate logistics routes. While spatial dynamics are modelled in detail, the study does not include analytical inventory prioritisation methods such as ABC×XYZ classification. As a result, SKU-level importance is not considered in routing or placement decisions.

Kofler et al. [3] explore affinity-based slotting strategies under dynamic order patterns. Their research highlights the importance of grouping frequently co-ordered items to reduce travel distances and improve picking efficiency. Although the approach incorporates order structure and demand interdependencies, it does not integrate real-time congestion data or spatial heat map analysis into routing decisions.

Winkelhaus and Grosse [4] present a systematic literature review of 114 articles on Logistics 4.0, deriving a framework that links external triggers (mass customisation, sustainability), digital technological building blocks (IoT, cyber-physical systems, Big Data, cloud, mobile and AR-based systems) and logistics tasks across supply, intra-, distribution and reverse domains. The review identifies warehousing and warehouse management as the single most represented task in the sample, with IoT-based WMS and real-time visibility repeatedly reported as drivers of improved order picking, inventory accuracy and material handling. However, the authors explicitly note that the field remains dominated by conceptual works and that the integration of analytical inventory prioritisation with the spatial dynamics of intralogistics is still underexplored — the exact gap addressed by the present work.

Stroumpoulis and Kopanaki [6] complement this picture at the strategic level. Their systematic review of sustainable supply chain management under digital transformation surveys how Big Data Analytics (BDA), IoT, and Blockchain technology (BT) are discussed in the academic literature in relation to economic, social, and environmental sustainability. The review reveals that these technologies are analysed almost exclusively through a strategic or conceptual lens: researchers examine their role in supply chain design, supplier selection, and triple-bottom-line reporting, but operational algorithmic frameworks for applying BDA or IoT-derived data within intralogistics processes are largely absent. This finding reinforces the gap identified by Winkelhaus and Grosse: while the potential of digital technologies is extensively theorised at the macro level, their translation into concrete routing and slotting optimisation algorithms remains an open research question — which the present work directly addresses.

The professional publication by LIDD Consultants [7] demonstrates the use of warehouse productivity heat maps for visualizing traffic intensity and identifying high-load zones. Heat maps provide valuable operational insights and support congestion analysis. Nevertheless, the approach is primarily descriptive and is not formally integrated with analytical SKU prioritisation or routing algorithms.

Overall, existing research addresses inventory prioritisation, spatial tracking, heat map visualisation, and routing optimisation as largely independent problems. Analytical methods such as ABC and XYZ classification provide demand-based prioritisation but lack spatial awareness. Conversely, RTLS-based tracking and heat map analysis capture movement dynamics but do not incorporate demand-driven importance of SKUs into decision-making models.

The large-scale literature review by Casella et al. [9] (2023) provides direct academic evidence that this separation is methodologically problematic. The study explicitly concludes that order picking operations should be optimized through the joint solution of a wide set of interdependent problems, and that sequential (isolated) optimisation of individual design tasks is likely to yield suboptimal performance. This finding strongly supports the integrated modelling approach adopted in the present work. Furthermore, Casella et al. document that picker blocking and aisle congestion lead to increased idle time, waiting periods, and overall efficiency losses — providing academic justification for introducing the congestion weight c_e into the routing cost function $\text{Cost}(e) = d_e + \lambda c_e$.

In the area of order picking routing, Ratliff and Rosenthal [11] (1983) provide the foundational result for single-block warehouses with return-to-depot routing. They prove that an optimal picking route in a rectangular warehouse with a single entry/exit can be computed in polynomial time using dynamic programming. This result establishes the theoretical basis for the routing setting adopted in the present study. Petersen [12] (1997) extends this line of work by comparing multiple routing heuristics in single-depot warehouses and evaluating their performance under different order sizes and layout configuration.

Practical industry evidence is provided by the GEODIS optimisation report [10], which documents a real-world warehouse transformation combining slotting reorganization and wave-based routing improvement. The report directly confirms the hypothesis underlying the present work: “good slotting without good routing still leads to suboptimal results, and good routing without good slotting means items are placed sub-optimally.” The GEODIS experience demonstrates that the true potential of warehouse optimisation lies in the integration of intelligent slotting and routing strategies, since together they maximize overall operational efficiency — which is precisely the objective of the 2×2 experimental design proposed in this study.

The absence of a unified framework that integrates ABC×XYZ analysis, congestion-aware routing, and spatial heat map data represents the main research gap addressed in this work. Although these approaches are widely used, they are usually applied separately.

Inventory classification methods focus on demand characteristics and economic importance of SKUs, but they do not consider warehouse layout or movement patterns. On the other hand, RTLS-based tracking and heat maps analyze spatial activity and congestion, but they do not include demand-based prioritisation. Routing models often minimise distance or time, yet they typically ignore both SKU importance and dynamic congestion.

To the best of the author’s knowledge, there is no clear methodological framework that combines these three components within a single optimisation model. As highlighted in the topic defence presentation, this separation limits the overall efficiency of warehouse optimisation, since

improving only one aspect does not guarantee improvement of the entire system.

This study aims to address this gap by developing an integrated graph-based framework that connects inventory prioritisation, congestion modelling, and routing decisions. The proposed approach allows joint evaluation of slotting and routing strategies and supports more comprehensive warehouse performance improvement.

3 Overview of the Proposed Method

This section presents the conceptual framework of the proposed approach to warehouse slotting and routing optimisation. The method combines inventory classification, spatial congestion modelling, and routing optimisation within a single integrated structure.

3.1 General Concept

The main objective of the proposed approach is to reduce both picking route length and aisle congestion by jointly considering demand-based SKU prioritisation and spatial warehouse activity. Route length is used as a proxy for picking time under the standard assumption that movement speed is roughly constant.

Traditional warehouse optimisation methods usually focus on only one component of the system. Inventory models prioritize items based on economic importance and demand variability, while spatial tools analyze movement patterns and congestion. In contrast, the proposed method connects these components and evaluates their interaction.

The framework is based on three main elements:

1. ABC×XYZ classification of SKUs,
2. Heat map construction based on RTLS/IPS movement data (or simulated tracks),
3. Congestion-aware routing within a graph-based warehouse model.

These elements are not applied independently. Instead, they are integrated into a unified decision-making model.

3.2 Warehouse as a Graph

The warehouse layout is represented as a graph $G = (V, E)$, where:

- V is the set of nodes representing aisle cells, intersections, and pick locations,
- E is the set of edges representing feasible movements between nodes.

Each edge has a base distance that reflects physical movement cost. In addition, congestion information derived from heat maps is associated with edges. This allows spatial activity to influence routing decisions. Such representation enables consistent modelling of both slotting and routing within the same structure.

3.3 Slotting Logic

At the first stage, SKUs are classified using ABC×XYZ analysis. The ABC component reflects economic importance, while the XYZ component captures demand stability.

High-priority items (for example, AX or AY groups) are placed in more accessible storage locations. Lower-priority items (such as CZ group) are assigned to less accessible areas.

In the extended version of the method, congestion levels are also considered during slotting. Storage locations are evaluated not only by their distance to the warehouse entrance, but also by local traffic intensity. This allows better distribution of picking activity across the warehouse.

3.4 Routing Logic

For each order, a picking route is constructed within the warehouse graph. The cost of movement along an edge is defined as:

$$\text{Cost} = \text{Distance} + \lambda \cdot \text{CongestionWeight},$$

where λ controls the influence of congestion on routing decisions.

If $\lambda = 0$, routing is based only on physical distance. If $\lambda > 0$, the model prefers less congested paths, even if they are slightly longer.

This formulation allows flexible balance between shortest-path efficiency and congestion avoidance.

3.5 Experimental Design

To evaluate the impact of different components, a 2×2 experimental design is used. Slotting strategies:

- ABC×XYZ slotting,
- ABC×XYZ + heat map slotting.

Routing strategies:

- Shortest path routing,
- Congestion-aware routing.

This results in four configurations:

1. ABC×XYZ + shortest path,
2. ABC×XYZ + congestion-aware routing,
3. ABC×XYZ + heat map slotting + shortest path,
4. ABC×XYZ + heat map slotting + congestion-aware routing.

Such structure makes it possible to evaluate separately the contribution of improved slotting and congestion-aware routing, as well as their combined effect.

3.6 Methodological Contribution

The key contribution of the proposed approach lies in the integration of analytical inventory prioritisation with spatial congestion modelling inside a unified graph-based framework. Instead of optimizing isolated components, the method evaluates how demand structure and movement dynamics influence each other.

This integrated perspective enables more comprehensive analysis of warehouse performance and provides a foundation for further formal modelling and experimental evaluation.

4 Formal Problem Formulation

In this section, the warehouse slotting and routing problem is described in a formal way. The goal is to define the main sets, decision variables, cost structure, and optimisation objective.

4.1 Basic Sets

Let:

- $S = \{s_1, s_2, \dots, s_n\}$ be the set of SKUs,
- $L = \{l_1, l_2, \dots, l_m\}$ be the set of storage locations,
- $O = \{o_1, o_2, \dots, o_K\}$ be the set of customer orders.

Each order $o_k \in O$ is a subset of SKUs:

$$o_k \subseteq S.$$

Each SKU s_i has a priority weight w_i derived from the ABC×XYZ classification. Higher values of w_i correspond to more important items.

4.2 Warehouse Graph

The warehouse layout is modelled as a graph:

$$G = (V, E),$$

where:

- V is the set of nodes (aisle cells, intersections, pick positions),
- E is the set of edges representing possible movements.

Each edge $e \in E$ has:

- a base distance $d_e > 0$,
- a congestion weight $c_e \geq 0$ derived from heat map data.

4.3 Slotting Decision

The slotting problem consists of assigning each SKU to a storage location. Define a binary decision variable:

$$x_{i,j} = \begin{cases} 1, & \text{if SKU } s_i \text{ is assigned to location } l_j, \\ 0, & \text{otherwise.} \end{cases}$$

Each SKU must be assigned to exactly one location:

$$\sum_{j=1}^m x_{i,j} = 1 \quad \forall s_i \in S.$$

Each storage location has limited capacity. For simplicity, assume capacity equal to 1:

$$\sum_{i=1}^n x_{i,j} \leq 1 \quad \forall l_j \in L.$$

4.4 Routing Cost

For each order o_k , a picking route is constructed in graph G . The cost of traversing edge e is defined as:

$$\text{Cost}(e) = d_e + \lambda \cdot c_e,$$

where $\lambda \geq 0$ controls the influence of congestion.

When $\lambda = 0$, routing depends only on physical distance. When $\lambda > 0$, congestion avoidance becomes part of the routing decision.

Let P_k denote the path chosen for order o_k . The total routing cost for this order is:

$$C_k = \sum_{e \in P_k} (d_e + \lambda \cdot c_e).$$

This formulation corresponds to a variant of the return routing problem [11], extended with congestion-aware edge weights controlled by λ .

4.5 Objective Function

The overall objective is to minimise the expected total picking cost over all orders. Let p_k denote the probability (or frequency) of order o_k . The objective function is:

$$\min \sum_{k=1}^K p_k \cdot C_k.$$

Since routing depends on SKU locations, and SKU locations depend on slotting decisions $x_{i,j}$, the routing cost is indirectly determined by slotting. Therefore, the problem integrates:

- slotting decisions $x_{i,j}$,
- congestion-aware routing,

- order demand structure.

4.6 Integrated Nature of the Problem

The complexity of the problem comes from the interaction between placement and routing. Changing SKU locations modifies pick nodes in the graph, which changes optimal paths and total cost.

The proposed framework does not aim to solve the global optimisation problem exactly. Instead, it evaluates heuristic slotting and routing strategies within this formal structure and compares their performance under different configurations.

5 Experimental Setup

This section describes the experimental environment used to evaluate the proposed slotting and routing framework. The implementation is based on a simulated warehouse model represented as a grid and a corresponding walkability graph.

5.1 Warehouse Layout Generation

The warehouse is modelled as a two-dimensional grid of size $R \times C$, which provides the spatial structure for the graph $G = (V, E)$ defined in the previous section.

Each cell of the grid belongs to one of three categories:

- 0 — aisle (walkable cell),
- 1 — storage location (shelf, non-walkable),
- 2 — door (walkable; used as both entrance and exit).

The set of storage locations $L = \{l_1, \dots, l_m\}$ corresponds to all grid cells marked as shelves (value 1). Each location $l_j \in L$ represents a potential assignment point for SKU $s_i \in S$ through decision variable $x_{i,j}$ defined in the formal model.

For experimental simplicity, all storage locations are assumed to have identical capacity. Each shelf cell can store at most one SKU unit, which corresponds to the capacity constraint

$$\sum_{i=1}^n x_{i,j} \leq 1, \quad \forall l_j \in L.$$

This assumption allows clear comparison of slotting strategies without introducing additional variability related to heterogeneous storage capacity.

Shelves are arranged in parallel rows separated by aisles of fixed width. Each shelf block has predefined dimensions (length and height), and shelf rows are separated by corridors to allow movement of picking agents. This structured layout ensures that all pick locations are accessible through walkable cells.

A single door is placed on the boundary of the grid and serves as both entry and exit point. In the graph model, the door corresponds to a specific node $v_{\text{door}} \in V$. All picking routes start and end at this node, which allows consistent computation of routing cost C_k for each order $o_k \in O$.

The single-door configuration with a combined entry and exit reflects the operational reality of small-to-medium warehouses, where receiving and dispatch typically share the same access infrastructure due to space, security, or cost constraints. This is common in regional 3PL facilities, e-commerce fulfillment centres, pharmaceutical and controlled-access storage, cold-chain warehouses (where each additional door increases heat loss), and buffer warehouses inside manufacturing plants. The resulting routing problem — where every picker starts at the depot, visits a set of shelf locations, and returns to the same depot — belongs to the return routing class formalised by Ratliff and Rosenthal [11].

With a grid resolution of approximately 1 m per cell, the 52×52 grid corresponds to a total floor area of roughly 2700 m^2 , which is representative of small-to-medium regional distribution facilities. The $|S| = 1000$ SKUs used in the experiment fall within the typical range for this segment (1000–10 000 SKUs). The generated layout contains $|L| = 1040$ shelf cells, which is slightly larger than $|S|$; the remaining 40 shelf cells stay empty in every slotting configuration, so the assignment $x_{i,j}$ is one-to-one on the placed SKUs.

The layout generation follows deterministic geometric rules to ensure:

- Regular shelf alignment,
- Uniform aisle width,
- Full connectivity of walkable cells.

The generated layout features a uniform aisle structure without a dedicated front cross-aisle near the depot. This design choice amplifies congestion concentration at the entry point and may underestimate the effectiveness of routing-only interventions in warehouses with more sophisticated aisle hierarchies (e.g., wide main aisles, front/back cross-aisles, one-way picking lanes). Validation on layouts with heterogeneous aisle widths is left for future work.

The resulting structure represents a simplified but realistic warehouse environment suitable for controlled experimental comparison of slotting decisions $x_{i,j}$ and routing strategies. In this model, each shelf cell represents a single pick location at the SKU level, abstracting away physical bin sharing. Although real shelves in medium-sized warehouses often hold 5–20 SKUs per bin, the unit-capacity assumption isolates the spatial effect of slotting and allows cleaner comparison between placement strategies.

5.2 Input SKU Dataset and ABC×XYZ Classification

The experiment uses a synthetic dataset consisting of $|S| = 1,000$ stock-keeping units (SKUs) with 12 months of historical demand data (January through December). Each record contains monthly demand figures, unit price, and total annual sales value. The dataset was sourced from the publicly available ABC-XYZ Inventory Classification Dataset published on Kaggle [8].

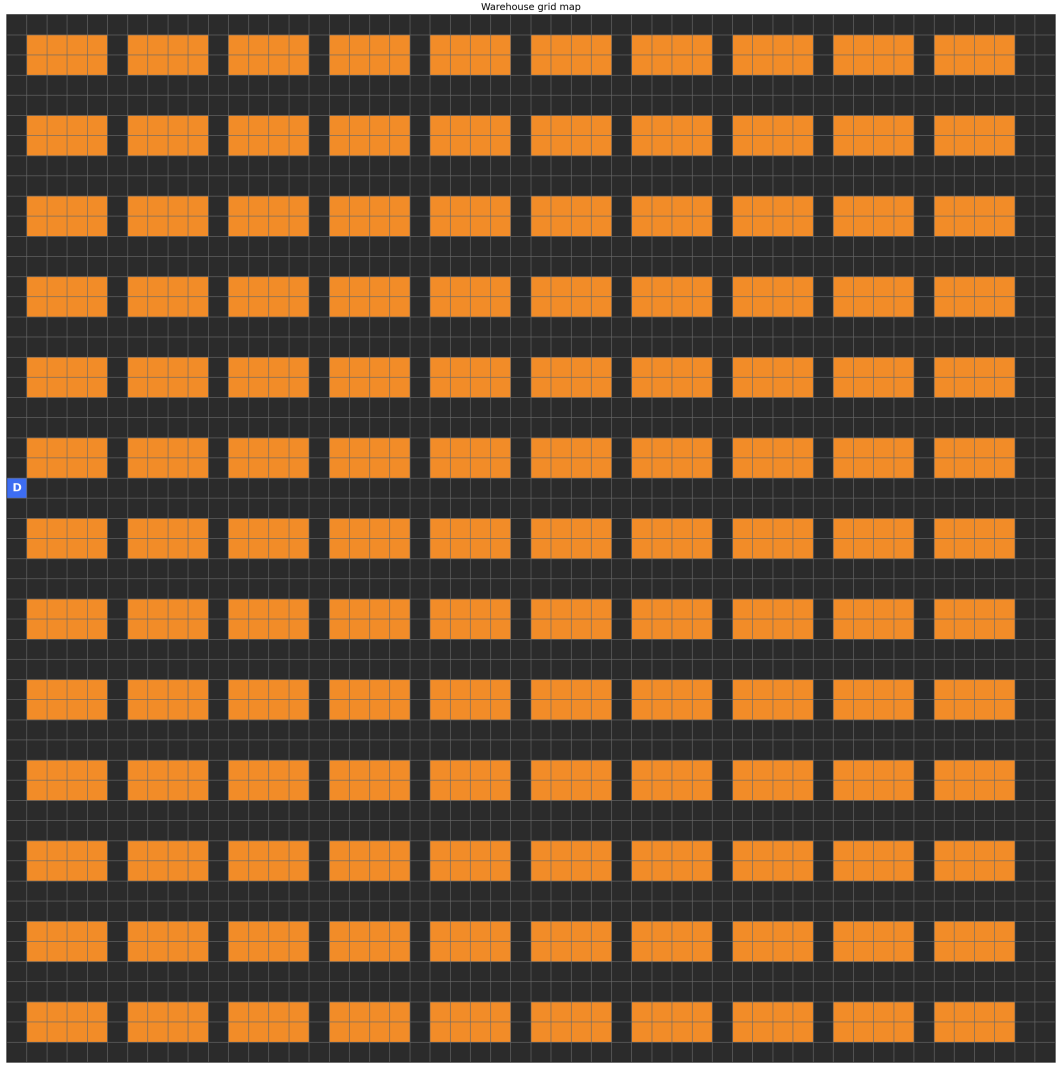


Рис. 1 – Generated warehouse layout ($R \times C = 52 \times 52$). Dark cells represent walkable aisles, orange cells represent storage locations (shelves), and the blue cell on the left boundary at row 23 indicates the single door node v_{door} .

This dataset was selected as the most suitable publicly available resource for ABC×XYZ inventory classification experiments. It provides exactly the data structure required by the combined analysis methodology: item-level sales values for ABC segmentation and monthly demand time series for XYZ variability assessment via the coefficient of variation. Unlike generic sales datasets, it was specifically designed to support supply chain segmentation tasks, making it directly applicable to the warehouse slotting context studied in this work.

ABC Classification. Items are ranked by descending total annual units (demand volume) and segmented into three groups following the APICS 70/20/10 convention by the number of items:

- A — top 20 % of items (200 SKUs) by annual demand volume — the highest-priority group placed closest to the door;
- B — next 30 % of items (300 SKUs) by annual demand volume;
- C — remaining 50 % of items (500 SKUs) by annual demand volume.

Because order sampling is weighted by total annual units, the resulting order probability distribution across classes is: A – 56.2%, B – 33.7%, C – 10.1%. This distribution is substantially more balanced than the revenue-weighted alternative (which concentrates 92.6% of order probability in class A), and reflects the reality that items with high unit volume are not always the same as items with high revenue value. In practice, item-count proportions frequently deviate from the 20/30/50 convention: Trubchenko et al. [1] report a 5/16/79 distribution in a real retail assortment, confirming that the thresholds define value boundaries, not item-count quotas. The synthetic dataset used here adheres to the imposed convention by construction.

XYZ Classification. Demand stability is quantified by the coefficient of variation $CV_i = \sigma_i/\mu_i$, where μ_i and σ_i are the mean and standard deviation of monthly demand for SKU s_i . Items are classified using thresholds following Trubchenko et al. [1]:

- X – $CV_i \leq 0.10$ (stable, predictable demand): 637 SKUs;
- Y – $0.10 < CV_i \leq 0.25$ (moderate variability): 216 SKUs;
- Z – $CV_i > 0.25$ (high variability, irregular demand): 147 SKUs.

The resulting 3×3 matrix of combined ABC×XYZ categories is shown in Figure 2. The mean coefficient of variation across all SKUs is $\overline{CV} = 0.143$, indicating that the dataset is predominantly stable. X-type items account for 63.7% of the total assortment, and the highest-priority sub-class AX contains 136 SKUs — these are positioned closest to the door in the baseline slotting strategy. With demand-unit-based order sampling, the baseline traffic Gini coefficient is 0.6697 (compared to 0.8214 under revenue-weighted sampling), confirming that traffic is now more evenly spread across the warehouse.

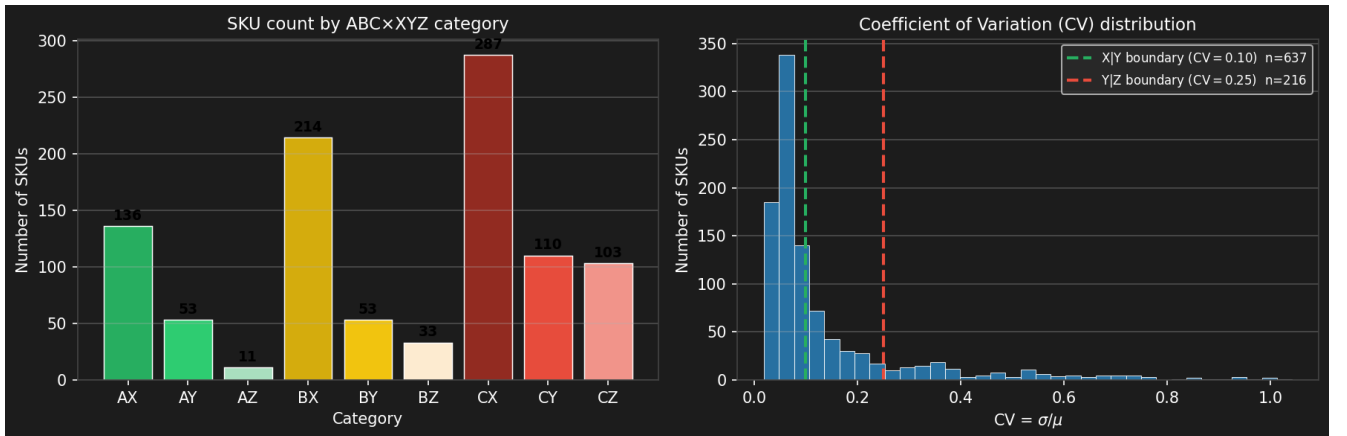


Рис. 2 – Left: number of SKUs per ABC×XYZ category. Right: distribution of the coefficient of variation CV with XYZ threshold boundaries ($CV = 0.10$ and $CV = 0.25$). X-type items dominate, reflecting stable seasonal demand patterns in the dataset.

5.3 Baseline Graph Model and Initial Configurations

After generating the grid, the warehouse is represented as a walkability graph

$$G = (V, E),$$

as defined in Section 4.

The node set V consists of all walkable grid cells (values 0 and 2), namely aisle cells and the door. The edge set E is defined using four-neighbor connectivity (up, down, left, right). An edge $e = (u, v) \in E$ exists if both cells are walkable and adjacent.

In the baseline configuration, the movement cost between adjacent nodes is defined in the simplest possible way. Each edge has unit weight:

$$d_e = 1.$$

Thus, the distance between two nodes corresponds to the number of movement steps required to travel between them. For any order $o_k \in O$, the routing cost under the baseline model is computed as

$$C_k = \sum_{e \in P_k} d_e,$$

where P_k is the selected path in graph G . Since $d_e = 1$ for all edges, the total cost is equal to the path length.

5.3.1 Baseline A: Classical ABC×XYZ Slotting

In Baseline A, slotting decisions $x_{i,j}$ are determined solely by ABC×XYZ classification.

SKUs $s_i \in S$ are ranked according to their combined category (AX, AY, ..., CZ) and economic importance. Storage locations $l_j \in L$ are ranked by their accessibility, measured as the shortest-path distance from the door node v_{door} to the nearest adjacent aisle cell.

Each SKU is assigned to exactly one shelf cell ($\sum_j x_{i,j} = 1$), and each shelf cell holds at most one SKU ($\sum_i x_{i,j} \leq 1$). This corresponds to the single-location assignment constraint from the formal model:

$$\sum_{j=1}^m x_{i,j} = 1, \quad \sum_{i=1}^n x_{i,j} \leq 1.$$

Importantly, congestion information is not used in this baseline. Placement decisions depend only on demand priority and geometric accessibility.

Figure 3 shows the distribution of BFS distances across all $|L| = 1,040$ shelf cells in the generated layout. The distribution is approximately unimodal with a mean distance of $\bar{\delta} = 38.9$ steps. Shelf cells closest to the door (distance ≤ 10 steps) correspond to high-priority AX and AY zones; cells at the far end of the range (distance ≥ 70 steps) are assigned to CZ items. The spread of the distribution reflects the geometric depth of the warehouse and confirms that the slotting mechanism creates a meaningful priority gradient from door to rear.

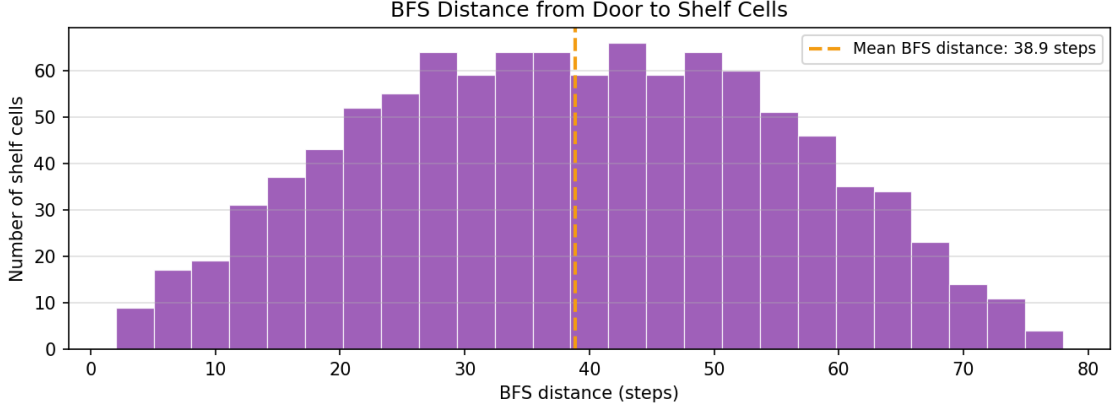


Рис. 3 – Distribution of BFS distances from the door node v_{door} (row 23, left boundary) to all $|L| = 1,040$ shelf cells in the 52×52 warehouse layout. The dashed line marks the mean distance $\bar{\delta} = 38.9$ steps. AX-priority items occupy the left tail (shortest distances), while CZ items are assigned to the right tail (longest distances).

5.3.2 Baseline B: Distance-Based Routing Without Congestion

Baseline B evaluates routing independently from congestion effects.

For each order $o_k \in \mathcal{O}$, the picking path is constructed using a shortest-path search on graph G with edge weights $d_e = 1$. No congestion penalties c_e are included in the cost function. Therefore, routing decisions are based purely on geometric distance.

Slotting configuration remains the same as in Baseline A. The only difference is the routing logic, which minimizes step count without considering traffic intensity.

5.3.3 Order Generation and Connection to Baselines

To evaluate Baseline A (slotting) and Baseline B (routing), a synthetic set of orders

$$\mathcal{O} = \{o_1, o_2, \dots, o_K\}$$

is generated using demand information from the ABC×XYZ dataset.

Let d_i denote the average monthly demand of SKU $s_i \in \mathcal{S}$. Each order is created by sampling a fixed number of distinct SKUs without replacement. The probability of selecting SKU s_i is proportional to its average demand:

$$p_i = \frac{d_i}{\sum_{j=1}^{|\mathcal{S}|} d_j}.$$

This design reflects a realistic assumption: items with higher demand are more likely to appear in customer orders. A fixed random seed is used to ensure reproducibility of the generated orders.

The generated orders connect the two baselines in the following way:

- Baseline A defines where each SKU is stored (the mapping from SKUs in $\mathcal{S}$ to shelf locations in L).

- Baseline B defines how a picker moves through the warehouse graph $G = (V, E)$ to collect all SKUs from an order o_k .

For a given order o_k , the picker starts at the door node v_{door} , visits the locations needed for the SKUs in o_k , and returns to the door. The routing cost is computed using the baseline edge weights $d_e = 1$, so the route cost is equal to the total number of movement steps:

$$C_k = \sum_{e \in P_k} d_e.$$

Since high-demand SKUs have larger probabilities p_i , they appear more often across orders. As a result, the corresponding shelf areas are visited more frequently. Even without explicit congestion modelling, this creates repeated traffic patterns near popular storage zones, which motivates the later introduction of heat maps and congestion-aware routing.

5.3.4 Order Picking Simulation in the Baseline Model

To illustrate how the model operates in practice, we describe the processing of a single order $o_k \in \mathcal{O}$ under the baseline configuration.

Step 1: Identification of storage cells. Using the slotting configuration defined in Baseline A, each SKU from the order o_k is mapped to its assigned shelf cell(s) in the set L . Since shelf cells are non-walkable, the picker cannot enter them directly.

Step 2: Selection of access points. For each required shelf cell, an adjacent walkable aisle cell is selected. Among the neighboring aisle cells, the one with the smallest distance to the door v_{door} is chosen. This ensures consistent distance-based accessibility evaluation.

Step 3: Route construction. The picker starts at the door node v_{door} , sequentially visits all selected access points, and finally returns to the door. The visiting order is determined by increasing distance from the door, which corresponds to the baseline distance-based logic.

Each movement segment is computed using a shortest-path search on the graph $G = (V, E)$ with unit edge weights $d_e = 1$. Therefore, the total picking cost for order o_k is equal to the total number of movement steps:

$$C_k = \sum_{e \in P_k} d_e.$$

Route Visualisation. Figure 4 illustrates an example of a picking route under the baseline configuration. The purple line represents the picker’s movement through aisle cells, and the numbered markers indicate the sequence of visited pick locations.

Observation on Traffic Concentration. Even in the absence of explicit congestion modelling, the visualisation reveals repeated use of specific aisle segments. Since high-demand SKUs are placed closer to the door and appear more frequently in orders, the corresponding storage zones are visited repeatedly.

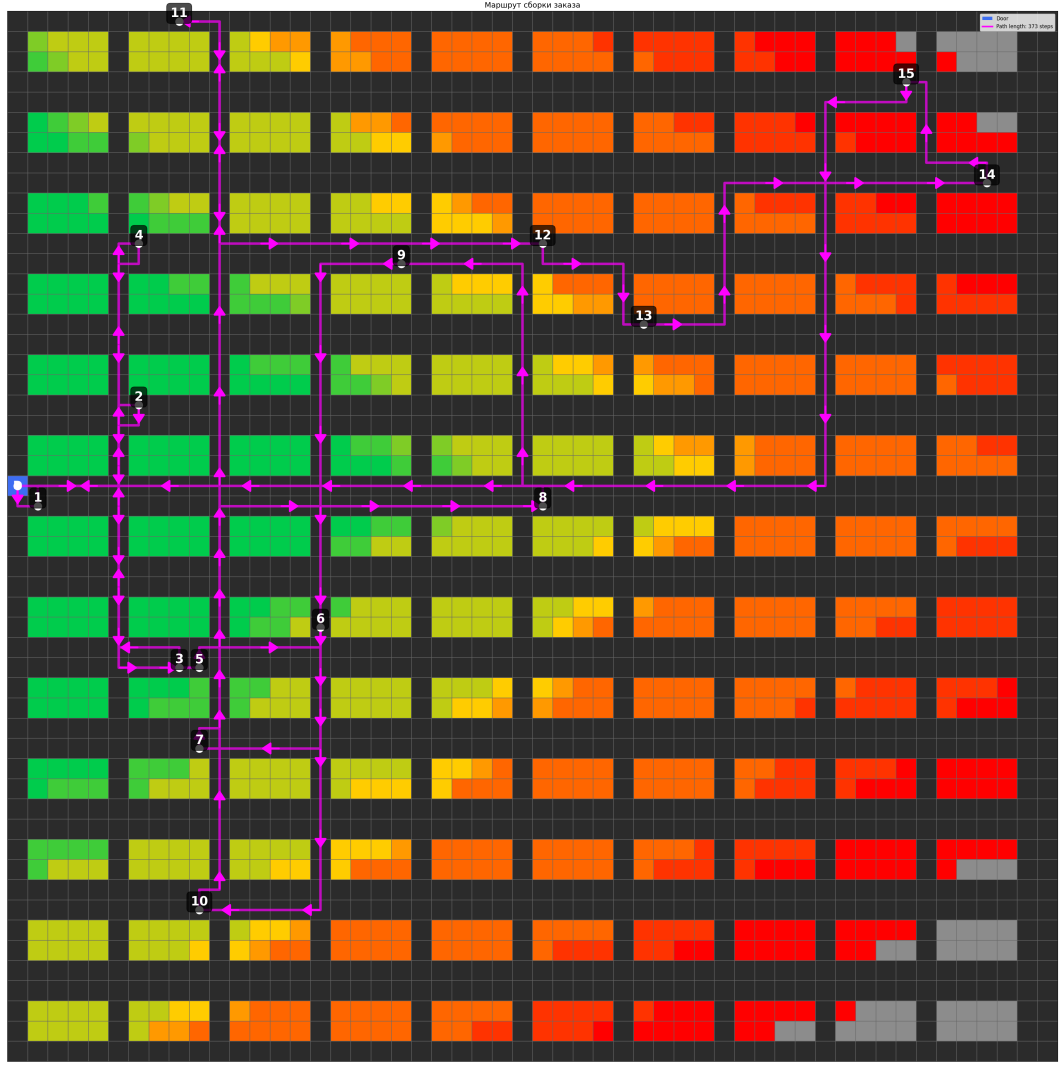


Рис. 4 – Example of a picking route under the baseline model. Shelf cells are coloured by ABC×XYZ priority (green = AX, red = CZ). The picker starts at the door node v_{door} , visits pick locations in order of increasing BFS distance, and returns to the door. Numbered markers indicate the visit sequence.

As a result, central corridors and areas near high-priority items experience higher traffic intensity. This effect motivates the introduction of heat map construction and congestion-aware routing in the extended model.

5.3.5 Resulting Baseline Configuration

The outcome of the two baseline components is:

- A slotting configuration where each SKU $s_i \in S$ is assigned to exactly one shelf cell $l_j \in L$ according to ABC×XYZ priority and BFS distance to the door (AX items closest, CZ items furthest);
- A routing model where all edge weights are $d_e = 1$, so the cost of each path equals the total number of movement steps, without any congestion penalty.

This configuration represents the simplest integrated warehouse model and serves as a reference point for evaluating congestion-aware extensions introduced in later sections.

5.4 2×2 Experimental Design

The 2×2 factorial design is used to measure the effect of each proposed improvement separately. Instead of testing one combined solution, the design treats two independent decisions — where items are stored (slotting) and how the picker moves between them (routing) — as separate factors. Crossing these two factors produces four distinct configurations. This structure allows the effect of each change to be measured on its own and also shows what happens when both changes are applied at the same time.

5.4.1 Design Variables

The two experimental factors are:

Slotting strategy (rows of the table).

- Classical slotting — ABC $\times$ XYZ placement without congestion awareness. High-priority AX/AY items are placed closest to the door based only on BFS distance.
- Heat-map-informed slotting — ABC $\times$ XYZ placement that also uses the observed congestion heat map. Items are moved away from zones that the baseline experiment identifies as structurally congested.

Routing strategy (columns of the table).

- Shortest-path routing ($\lambda = 0$) — Dijkstra / BFS with unit edge weights $d_e = 1$; no congestion penalty is used.
- Congestion-aware routing ($\lambda > 0$) — Dijkstra with the combined edge cost $\text{Cost}(e) = d_e + \lambda c_e$, which penalises heavily used corridors and encourages detours around congested areas.

5.4.2 Four Experimental Configurations

Crossing the two factors yields four distinct configurations (Table 1):

Таблица 1 – 2×2 experimental design: four warehouse configurations

	Shortest-path routing	Congestion-aware routing
Classical slotting	Baseline	Config A
Heat-map slotting	Config B	Config C (full integration)

- Baseline — ABC $\times$ XYZ slotting with shortest-path routing ($\lambda = 0$). This is the reference configuration described in the previous section. All performance improvements are measured relative to it.

- Config A (routing improvement only) — classical slotting with congestion-aware routing. Item placement does not change, but the algorithm penalises congested edges. Structural bottlenecks near the door are expected to persist, since slotting is not modified.
- Config B (slotting improvement only) — heat-map-informed placement with shortest-path routing. Moving high-demand items away from the congested entry zone targets the main cause of traffic concentration.
- Config C (full integration) — heat-map slotting combined with congestion-aware routing. When item placement already reduces structural bottlenecks, the routing algorithm has more room to choose alternative corridors. The combination of both axes is expected to give the largest overall gain.

The pre-experiment hypothesis is that the gains grow from Config A to Config C, but the exact magnitude depends on how the two axes interact. The experimental sections (Sections 9 and 10) report the measured values.

5.4.3 Evaluation Metrics

Four metrics are used to evaluate each configuration:

1. Weighted Path Congestion (WPC) — the main congestion metric. For a picking path P :

$$\text{WPC}(P) = \sum_{e \in E(P)} c_e,$$

where $c_e \in [0, 1]$ is the normalised congestion weight of edge e taken from the heat map. WPC shows how much congested corridor capacity a picking route uses. A lower WPC means the route avoids busy segments, which reduces the chance of picker delays even if the physical path is slightly longer.

2. Mean route length (steps) — the average number of movement steps needed to complete an order over all $K = 10,000$ orders:

$$\bar{L} = \frac{1}{K} \sum_{k=1}^K |P_k|.$$

This metric reflects picking time when movement speed is assumed to be constant. It is used as a proxy for order fulfilment efficiency.

3. 95th-percentile route length (L_{95}) — a robustness indicator based on the distribution of per-order route lengths:

$$L_{95} = \text{Percentile}_{95}(\{|P_k|\}_{k=1}^K).$$

While $\bar{L}$ shows average performance, L_{95} shows whether a configuration produces very long routes for a small but notable fraction of orders. A decrease in L_{95} across configurations means the improvement is consistent and not limited to simple orders.

Note on interpretation. In this model, congestion weights c_e are pre-computed from the baseline simulation and stay fixed during the experiment (single-picker, sequential routing). Therefore, L_{95} captures worst-case order compositions (e.g. orders whose items are in distant zones) rather than real-time congestion between pickers working at the same time. A dynamic, multi-picker simulation would be needed to study the latter, and this is left for future work.

Note on data partitioning. A train/test split is not used in this work. All $K = 10000$ orders are generated from the same fixed demand distribution and the same random seed, so every order is an independent sample drawn with identical probabilities. Building the heat map and evaluating routing on the same set therefore does not introduce a selection bias: a held-out subsample would have the same expected distribution as the training one. A more elaborate protocol with multiple independent order samples (for example, k -fold averaging over different seeds) is left for future work.

Together, these three metrics give a complete picture of the 2×2 design: WPC measures congestion load, $\bar{L}$ measures average efficiency, and L_{95} checks that improvements hold even for the most demanding orders.

5.5 Implementation and Volumetric Characteristics

All experiments were implemented in Python 3.12 and run on a MacBook Pro with an Apple M4 processor and 48 GB of RAM. The full source code base contains approximately 5500 lines of Python distributed across 28 files (about 270 KB of source text). The core of the implementation is the project-specific library `warehouse/`, which provides ten modules (`layout`, `graph`, `slotting`, `routing`, `model`, `metrics`, `viz`, `constants`, `__init__`, and a few helpers). The library accounts for approximately 2500 lines and 250 KB. The remaining files are experiment-specific scripts (sensitivity sweeps, plot generation, results dump) and three Jupyter notebooks (`thesis_notebook.ipynb`, `demo.ipynb`, and `test.ipynb`) which were used for interactive exploration and for the iterative construction of the heat maps and figures presented in the report.

The experimental volume is summarised as follows:

- Dataset: 1000 SKUs with 12 months of demand history, sourced from the public Kaggle ABC-XYZ Classification Dataset [8].
- Warehouse: 52×52 grid, 1040 shelf cells, and 1532 walkable cells.
- Orders: 10000 orders of 15 items each, sampled with demand-weighted probabilities under a fixed seed of 42.
- Slotting strategies tested: 7 in total — Baseline ABC $\times$ XYZ, DWC, HZR, TAPS, Aisle-Balanced, AAHMS, and the exploratory CAS.
- Routing algorithms tested: 2 (BFS and Dijkstra-NN); full $7 \times 2 = 14$ slotting–routing combinations.

- Sensitivity sweeps: $\Delta \in \{5, 10, 15, 20, 30, 50\}$; $h \in \{0.05, \dots, 0.30\}$ (6 values); $\alpha \in \{0, 0.5, 1, 2, 3, 5, 10\}$; $f_A \in \{0.10, \dots, 0.40\}$ (6 values); $K \in \{25, 50, 100, 150, 250, 400\}$; $\lambda \in \{1, 2, 5, 6, 7, 8, 10, 20\}$ on Baseline; $\lambda \in \{0, 1, 2, 5, 10, 20\}$ on CAS. Together these add 45 extra configurations.

A single full evaluation (one slotting combined with one routing algorithm, across $K = 10000$ orders) takes about 30–60 seconds on the M4 machine. Running the full 7×2 matrix together with the λ -sweep on CAS therefore takes roughly 40–45 minutes end-to-end. Peak resident memory stays below 1 GB; the dominant data structures (the visit-count field on 52×52 cells, the 10000-order list, and the per-strategy heat maps) all fit comfortably into RAM. All reported results use the fixed seed of 42, so every experiment is reproducible from the published scripts and notebooks.

6 Congestion-Aware Routing: Experiment and Results

6.1 Config A: Congestion-Aware Routing with Classical Slotting

This section evaluates Config A from the 2×2 experimental design: classical ABC $\times$ XYZ slotting (unchanged from the Baseline) combined with congestion-aware routing ($\lambda > 0$). Config A isolates the effect of routing alone: item placement stays the same, and only the navigation strategy changes.

The key question is: does replacing shortest-path routing with a congestion-penalised Dijkstra search reduce traffic on busy corridors, given that high-demand items are still clustered near the door?

The simulation uses the same 52×52 warehouse grid described in Section 5: 1000 SKUs in shelf cells, door at position (23,0), ABC $\times$ XYZ slotting with thresholds $A = 20\%$, $B = 30\%$, $X_{CV} = 0.10$, $Y_{CV} = 0.25$.

6.1.1 Baseline Traffic Heat Map

Before congestion-aware routing can be applied, the traffic pattern generated by the standard ABC $\times$ XYZ slotting must be recorded. A simulation of $K = 10000$ orders with BFS routing ($\lambda = 0$) is run, and the number of times each corridor cell is visited is counted. The result is the baseline traffic heat map $H : V \rightarrow \mathbb{N}$ shown in Figure 5.

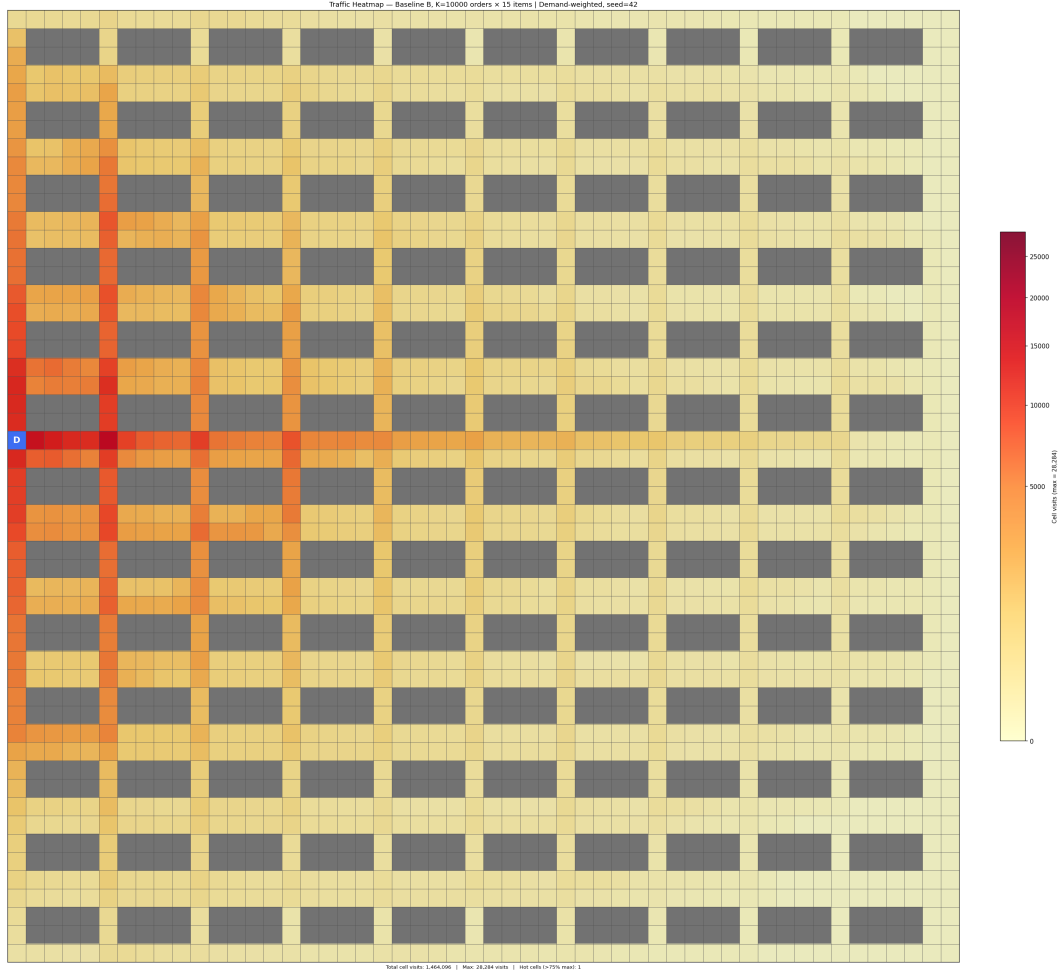


Рис. 5 – Baseline traffic heat map ($K = 10000$ orders, BFS routing, demand-unit-weighted). Cell colour indicates the number of picker traversals: dark red is the highest load, light yellow is low load, grey cells are shelf blocks. The total number of cell visits is 3 398 472; the maximum single-cell count is 35 158.

Traffic is extremely concentrated in the upper-left corner of the warehouse, near the door. The leftmost vertical aisle and the horizontal corridor at door height act as the primary bottleneck: every picker must pass through them to reach the A-class shelves clustered in that zone. The rest of the warehouse — roughly the right two-thirds — is barely used, with cell visit counts close to zero.

6.1.2 Edge Congestion Weights c_e

The congestion field for Config A is built directly from the $K = 10000$ generated orders. Each walkable edge $e = (u, v) \in E$ collects a traversal count, which is then normalised by the warehouse-wide maximum:

$$c_e = \frac{\text{count}(e)}{\max_{e' \in E} \text{count}(e')}, \quad c_e \in [0, 1].$$

Edges never traversed during the simulation receive $c_e = 0$. This congestion map is fixed and used as the static weight input to Config A; all reported metrics (mean steps, L_{95} , WPC) are computed

on the same $K = 10000$ orders. As explained in the note on data partitioning in Section 5.5, a train/test split is not used here because every order is an independent sample drawn from the same fixed distribution.

6.1.3 Congestion-Aware Routing Algorithm

The Baseline routes each order with a simple distance-sorted + BFS pipeline. Target shelves are visited in increasing order of BFS distance from the door, and the segment between two consecutive stops is found by an unweighted BFS. This is a deliberately minimal routing model that uses no clever visit-order heuristic; it serves as the simplest possible benchmark against which the value of congestion-aware routing can be measured.

In Config A, this pipeline is replaced by a single congestion-aware mechanism that decides both which stop to visit next and how to reach it, using the same edge cost everywhere:

$$\text{Cost}(e) = 1 + \lambda \cdot c_e, \quad \lambda = 5.0,$$

where $c_e \in [0, 1]$ is the pre-computed normalised congestion weight of edge e . With $\lambda = 5$, a maximally congested edge is penalised as heavily as five additional movement steps, which strongly encourages the algorithm to take detours around hot spots.

The routing procedure is the Dijkstra-driven nearest-neighbour heuristic (Dijkstra-NN):

1. At the picker’s current position p (initially the door), run a single-source Dijkstra with the cost above. In one pass this returns the cost-distance from p to every cell in the warehouse.
2. Among the remaining unvisited stops, pick the one with the lowest Dijkstra cost from p .
3. Rebuild the path to that stop from the Dijkstra predecessor map, append it to the route, and update p to the new position.
4. Repeat steps 1–3 until every stop has been visited; finally compute one more Dijkstra from the last stop back to the door.

The key property of Dijkstra-NN is metric consistency: the same $1 + \lambda c_e$ cost decides both the visit order (greedy by Dijkstra distance from the current position) and the path between consecutive stops. A static sort by door distance — the equivalent of the Baseline’s visit order — would use one metric in the outer loop and a different metric in the inner Dijkstra, and could send the picker through a hot corridor on a “shorter-by-door” detour. Dijkstra-NN aligns the two metrics and recomputes the order after every visit, so that congestion penalties affect which item is picked next, not just how the picker reaches an item that has already been chosen.

On the choice of visit-order heuristic. Ratliff and Rosenthal [11] present a polynomial-time dynamic-programming algorithm for the joint visit-order-and-path problem, but only under strict topological assumptions: a single-block rectangular warehouse with parallel aisles and a fixed depot. The 52×52 grid used here does not meet those assumptions: shelf rows are arbitrary, the

depot lies on the perimeter, and paths may take any shape on the full adjacency graph. R&R’s DP is therefore not directly applicable in this study. The survey by Petersen [12] compares single-depot routing heuristics (S-shape, largest-gap, NN) and reports that NN is typically within 10–15% of structured strategies on rectangular layouts. Because of the topology gap above, neither S-shape nor largest-gap can be used on the present grid without modification, so the comparison in this work is restricted to Baseline (distance-sorted) versus Config A (Dijkstra-NN).

Route length alone does not show the quality of a path in terms of congestion: a longer route through empty corridors may be better than a shorter route through a hot spot. The Weighted Path Congestion (WPC) metric, defined in Section 7, captures this trade-off directly.

Table 2 shows route lengths and WPC values for both the Baseline and Config A for two representative orders discussed in the following subsections.

Таблица 2 – Route length and Weighted Path Congestion for two orders

Order	Routing	Steps	Δ Steps	WPC	Δ WPC
Order o_0 (random)	Baseline (distance-sort + BFS)	361	—	108.58	—
Order o_0 (random)	Config A (Dijkstra-NN, $\lambda = 5$)	249	−31.0%	24.89	−77.1%
Typical order	Baseline (distance-sort + BFS)	147	—	55.86	—
Typical order	Config A (Dijkstra-NN, $\lambda = 5$)	85	−42.2%	16.79	−69.9%

6.2 Representative Orders

The analysis uses two representative orders that cover opposite ends of the demand spectrum:

- Order o_0 (randomly selected) — the first order generated by the simulation. Its 20 items are drawn from the full demand distribution and span several shelf zones, making it a typical example of a spatially dispersed order.
- Typical order — a synthetic order composed of the 15 SKUs with the highest demand frequency across all 10 000 simulated orders. Because these items belong to the A-class and are stored in the zone directly adjacent to the door, this order represents the most common traffic pattern in the warehouse and puts maximum stress on the entry corridor.

Figure 6 shows the four picking routes: Baseline (BFS) and Config A (congestion-aware, $\lambda = 5.0$) applied to each order. Shelf cells are coloured by ABC×XYZ priority (green = AX/high-demand, orange = CZ/low-demand). The magenta line is the picking route; the letter D marks the door.

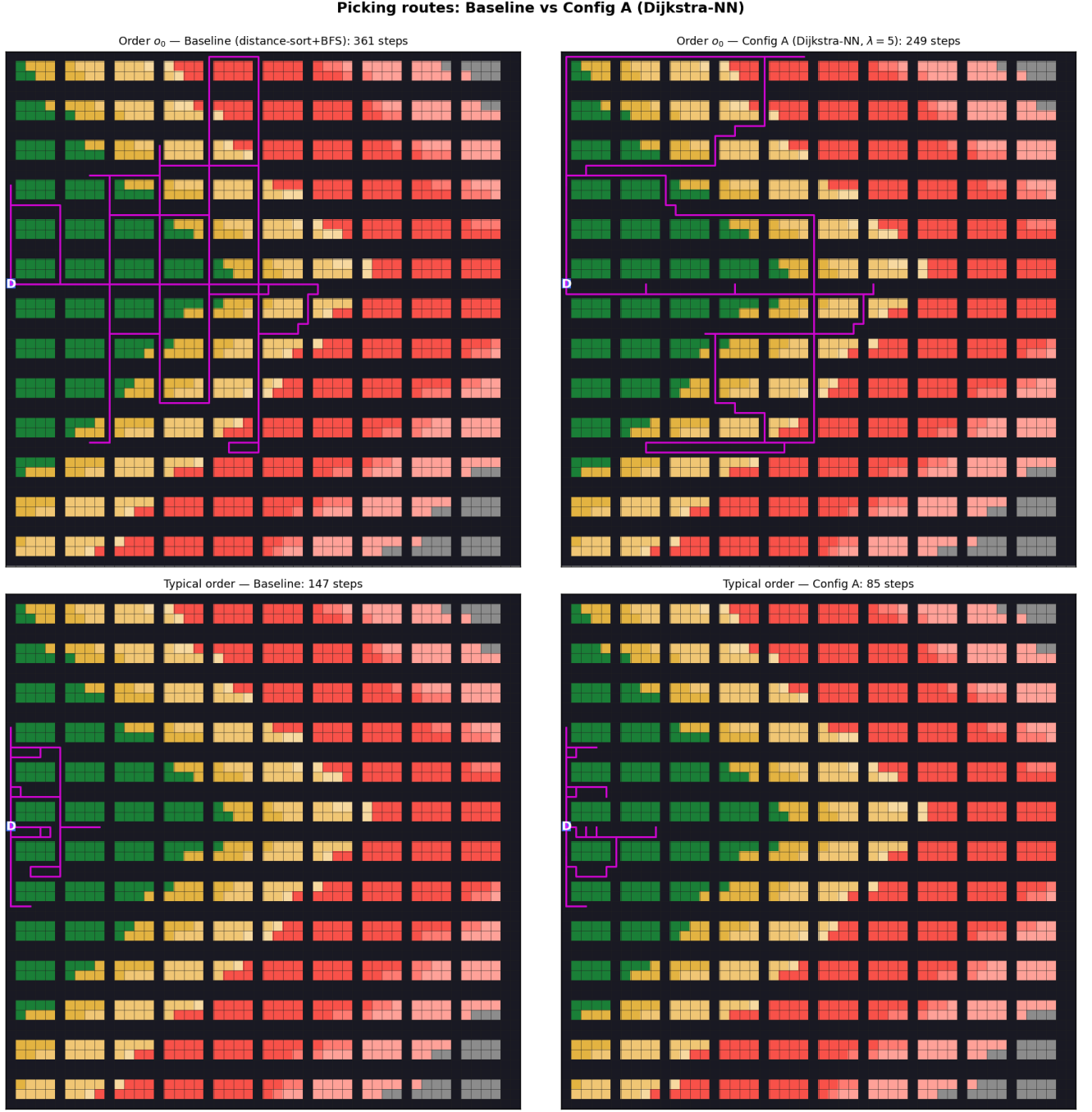


FIG. 6 – Picking routes for the two representative orders under BFS (left column) and congestion-aware routing with $\lambda = 5.0$ (right column). Top row: order o_0 (random). Bottom row: typical order (top-15 most frequent items). Shelf cells are coloured by ABC $\times$ XYZ category. The magenta line shows the picking path; D marks the door.

To complement the route visualisations, Figures 7 and 8 present a zone analysis of each order: every walkable cell is classified by which routing algorithms use it. Green cells are used by the Baseline but avoided by Config A (relieved by CA); blue cells are CA-only detours; red cells are shared by both routes (bottleneck). This view makes explicit where the Dijkstra-NN router achieves its savings.

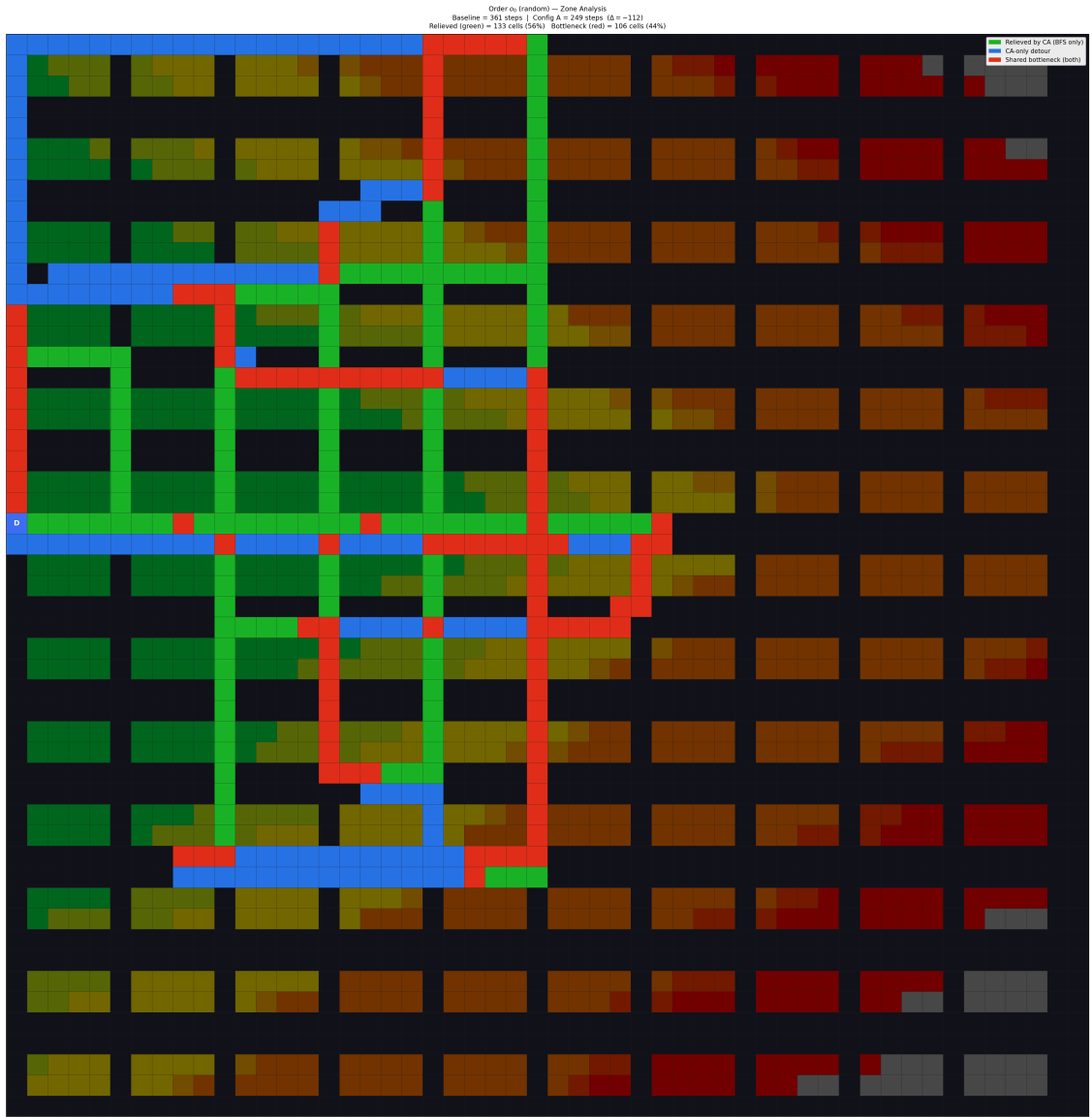


Рис. 7 – Zone analysis for order o_0 (random). Baseline route = 361 steps, Config A route = 249 steps ($\Delta = -112$). A substantial fraction of Baseline-only cells are relieved by CA; a shared bottleneck remains along the entry corridor that every A-class pick must cross. Shelves are tinted by ABC×XYZ category (dark green AX $\rightarrow$ dark red CZ) for spatial reference.

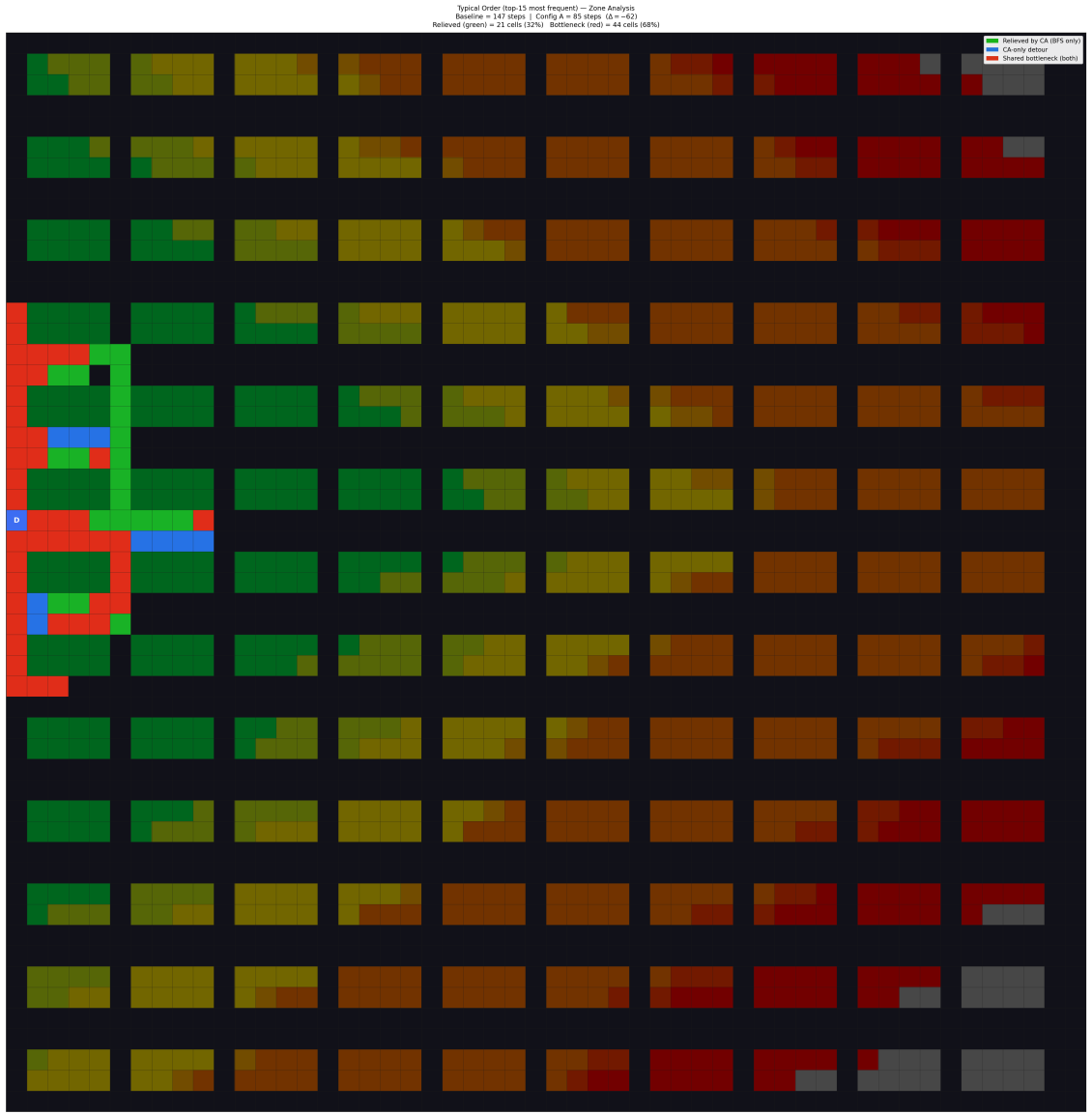


Рис. 8 – Zone analysis for the typical order (top-15 most frequent SKUs). Baseline route = 147 steps, Config A route = 85 steps ($\Delta = -62$). A portion of corridor cells are relieved while the door-adjacent edges shared by every A-class pick remain a shared bottleneck. CA still finds alternative paths that shorten the route by 62 steps relative to the Baseline.

6.3 Results

Random order o_0 . On the randomly selected order, Config A reduces the route from 361 to 249 steps (-31.0%) and the WPC from 108.58 to 24.89 (-77.1%). The metric consistency of Dijkstra-NN is the key factor: because both the visit-order and the path-finding use the same $1 + \lambda c_e$ cost, detours around congested edges are only taken when they also shorten the door-relative cost. The algorithm never chooses a clean detour that happens to be longer in practice than the cheapest non-detour.

Typical order. On the typical order — the 15 most frequently requested SKUs, all from the A-class zone next to the door — Config A reduces the route from 147 to 85 steps (-42.2%) and the WPC from 55.86 to 16.79 (-69.9%). The improvement is structurally limited: every pick in

the typical order must still cross the entry corridor at least once, so the remaining WPC of 16.79 reflects the unavoidable load on the door-adjacent edges themselves and not any weakness of the router. This remaining load is exactly the part that only a change in slotting can remove, which is the motivation for Section 9.3.

Aggregate metrics over $K = 10000$ orders. Across the full order set, Config A produces routes that are on average 34.7% shorter than the distance-sorted BFS Baseline; L_{95} falls from 439 to 281 steps (−36.0%). Mean WPC drops from 108.50 to 24.57 (−77.4%). Both operational criteria (see Section 9.3.6) are therefore satisfied by routing alone: the entry corridor is still the busiest single edge in the warehouse, but the average picker crosses it on a path that is shorter and less overloaded than the BFS reference. Table 3 gives the full numerical summary.

Таблица 3 – Aggregate metrics: Baseline vs Config A ($K = 10000$ orders, seed 42, $\lambda = 5.0$). Both routers use the same heat map built from the Baseline traffic.

Config	Mean L	L_{95}	Mean WPC
Baseline (distance-sort + BFS)	339.8 steps	439 steps	108.50
Config A (Dijkstra-NN, $\lambda = 5.0$)	221.9 steps	281 steps	24.57
Δ (Config A – Baseline)	−117.9 (−34.7%)	−158 (−36.0%)	−83.93 (−77.4%)

6.4 Discussion

Config A gives a clear improvement over the Baseline routing on both criteria at the same time: routes get shorter and WPC drops by roughly three quarters. Two points follow from the result.

1. Metric consistency is the key design choice. The Baseline router sorts stops by static door distance and uses BFS between them; Config A uses the same $1 + \lambda c_e$ cost for both ordering and path-finding (Dijkstra-NN). Once these two metrics agree, the algorithm no longer takes congestion detours that look short under the outer loop’s ranking but long under the inner loop’s. That mismatch is what creates pointless backtracking and inflates WPC without making the route shorter.
2. Routing alone cannot remove the structural bottleneck. Even with a −77.4% drop in mean WPC, the remaining congestion on the typical order (16.79) comes mostly from the door-adjacent edges that every A-class pick must cross at least once. With demand-unit-based order sampling, class A items account for 56.2% of order probability mass, and the remaining B/C traffic reaches further shelves — yet the entry corridor is on virtually every route regardless of how well the picker chooses the next stops. Routing lowers the average traversal cost of that corridor but cannot remove it.

Implications for the 2×2 design. Config A shows that the routing axis on its own makes a substantial contribution: −34.7% mean steps and −77.4% mean WPC, simply by replacing the

visit-order heuristic with one that shares the metric of the path-finder. The numbers in Table 3 set a strong reference against which the slotting changes (Configs B and C, Section 9.3) are judged. A slotting change only counts as a real improvement if it brings L_{95} or mean WPC below the 281 steps and 24.57 figures established here.

Sensitivity analysis for λ . The penalty weight $\lambda = 5$ was chosen as a representative operating point with moderate-to-strong detour pressure: a maximally congested edge carries the same cost as five additional movement steps. To check that the qualitative conclusion does not depend on this exact choice, λ was swept over $\{1, 2, 5, 6, 7, 8, 10, 20\}$. Table 4 reports the aggregate metrics for all $K = 10000$ orders together with the WPC of the typical order.

Таблица 4 – Congestion-aware routing sensitivity to $\lambda \in \{1, 2, 5, 6, 7, 8, 10, 20\}$. $K = 10000$ orders, seed 42, Dijkstra-NN throughout. Δ_L and Δ_{WPC} are relative changes vs. Baseline ($\lambda = 0$).

λ	Mean L (steps)	Δ_L	L_{95} (steps)	Mean WPC	Δ_{WPC}
Baseline (distance-sort + BFS)	339.8	—	439	108.5	—
1	218.0	−36%	279	29.6	−73%
2	218.5	−36%	279	27.4	−75%
5	221.9	− 35%	281	24.6	− 77%
6	223.3	−34%	281	24.1	−78%
7	224.7	−34%	283	23.7	−78%
8	226.1	−33%	283	23.4	−78%
10	229.5	−32%	287	22.9	−79%
20	252.1	−26%	321	21.1	−81%

The sweep shows a clean monotone trade-off rather than a scissors crossover. Every $\lambda > 0$ tested produces routes that are shorter and less congested than the Baseline, simultaneously, with no degradation regime at either extreme. Two observations.

Route length is non-monotone. It is minimised near $\lambda = 1$ (218.0 steps, −36%) and grows slowly with λ thereafter. This reflects the trade-off the parameter encodes: larger λ buys more detour at the price of slightly longer routes. Even at $\lambda = 5$, routes are 35% shorter than Baseline (221.9 steps vs. 339.8).

Mean WPC is monotone decreasing. It falls from 29.6 at $\lambda = 1$ (−73%) to 24.6 at $\lambda = 5$ (−77%) and continues to 21.1 at $\lambda = 20$ (−81%). Each increment of λ buys a few more avoided hot-edge traversals, with diminishing returns above $\lambda \approx 5$.

The selected value $\lambda = 5$ sits at the elbow of both curves: WPC has captured −77% of the available reduction by $\lambda = 5$, and the additional 4 extra steps relative to the route-length minimum at $\lambda = 1$ are a small price for the extra robustness against heatmap noise — at $\lambda = 1$, a 10% error in any c_e would change the detour decision; at $\lambda = 5$, the same error is buried in a much stronger congestion signal. At $\lambda = 20$, routes grow to 252.1 steps (−26%) while WPC reaches 21.1

(−81%) — still below baseline on both axes, confirming the absence of a crossover regime at any tested λ .

Figure 9 illustrates the curves; the $\lambda = 5$ operating point is marked. Unlike the original two-metric routing model where increasing λ helped one metric at the cost of the other (a true scissors pattern), Dijkstra-NN eliminates the contradiction: all positive λ values dominate the Baseline on both axes.

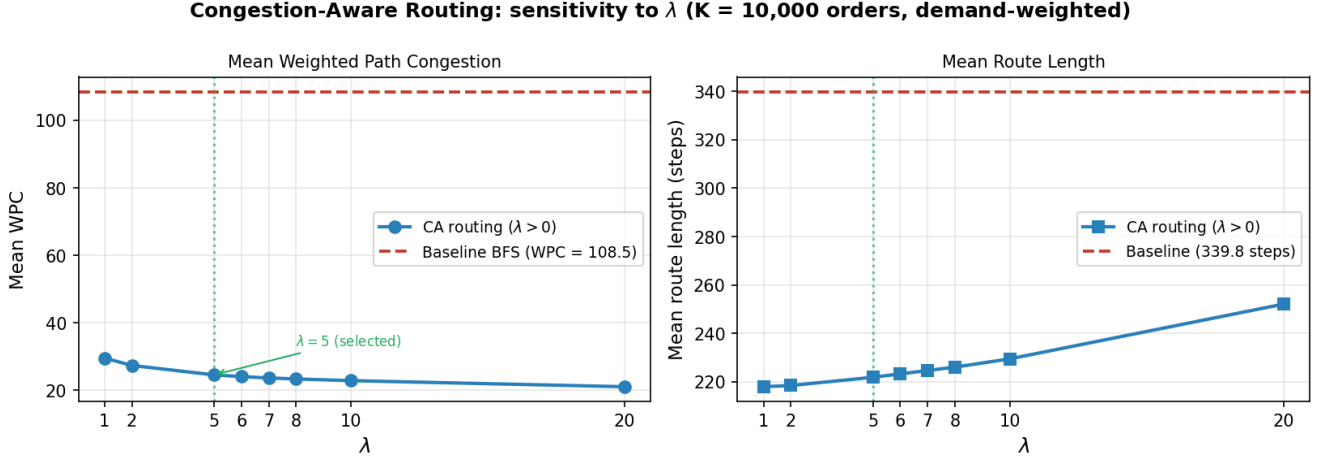


Рис. 9 – Congestion-aware routing sensitivity to $\lambda \in \{1, 2, 5, 6, 7, 8, 10, 20\}$ under Dijkstra-NN (demand-unit weighted). Left: mean WPC over all $K = 10000$ orders (dashed line = Baseline 108.5). Right: mean route length in steps (dashed line = Baseline 339.8 steps). Both metrics are below baseline at every confirmed $\lambda > 0$; the selected value $\lambda = 5$ sits at the elbow of both curves.

7 Heat-Map-Informed Slotting: Experiment and Results

7.1 Config B: Heat-Map Slotting with Shortest-Path Routing

7.1.1 Motivation

In the Baseline model, the most popular items — such as AX-class SKUs — are placed as close to the door as possible. This is the standard ABC×XYZ slotting rule: high-demand items should have short picking paths. In practice, however, it creates a structural problem. Because virtually all high-frequency orders start from the same door and immediately enter the same narrow zone, the central entry corridors become severely overloaded. The traffic heat map from Section 6 makes this visible: a small cluster of corridor cells near the door receives an order of magnitude more traversals than the rest of the warehouse.

Config A (Section 8) attempted to fix this at the routing level, but the congestion-aware algorithm cannot bypass a bottleneck when the bottleneck is the only gateway to the items. The root cause is the slotting decision, not the routing logic.

7.1.2 Heat-Map-Based Slotting Strategy

Config B changes the slotting algorithm instead of the routing algorithm. The routing stays the same as in the Baseline (BFS, $\lambda = 0$); only item placement is modified.

The key idea is that shelf attractiveness is defined not only by distance from the door, but also by the congestion intensity of the corridors around that shelf. A shelf that is slightly further from the door but reachable through a quiet corridor can be a better placement for a popular item than a closer shelf that requires passing through the most congested aisle.

7.1.3 Iterative Algorithm

The Config B placement procedure works as follows:

1. Build a baseline heat map. Starting from the standard ABC×XYZ slotting, run K orders with BFS routing and record the number of times each corridor cell is visited. This produces the traffic heat map $H : V \rightarrow \mathbb{N}$.
2. Identify overloaded zones. Select the top $p\%$ of corridor cells by visit count (for example, $p = 20$). These are the “hot zones” where congestion is highest. Shelves whose access cells fall within a hot zone are marked as candidates for relocation.
3. Relocate high-demand items. High-demand SKUs (A-class) that are currently placed in hot-zone shelves are moved to quieter but still accessible locations (“cold-zone” shelves). Cold-zone candidates are ranked by corridor congestion ascending and by BFS distance ascending, so that the algorithm prefers shelves that are both quiet and close to the door.
4. Rebuild the heat map and repeat. After relocation, the heat map is recomputed to reflect the new traffic pattern. The process can be repeated for two or three iterations until convergence.

The baseline traffic heat map used as the starting point for the slotting analysis is the same one constructed in Section 6.1.1: $K = 10000$ orders routed under standard ABC×XYZ slotting with BFS, yielding the cell-visit map $H : V \rightarrow \mathbb{N}$ shown in Figure 5. That heat map reveals the structural problem that motivates the slotting strategies evaluated below.

7.2 Identifying Overloaded Zones

To determine which shelves are candidates for relocation, the top $p = 20\%$ of corridor cells by visit count are selected. These are the “hot zones” where traffic intensity is highest. Within this group, the top 20% of the top-20% — that is, the top 4% of all visited cells — are classified as critical zones: locations so severely overloaded that any routing strategy is almost certain to pass through them.

In the baseline simulation, the hot-zone threshold is 1171 visits per cell, covering 307 corridor cells (20% of the 1532 active corridor cells). The critical-zone threshold is higher still, covering the 62 most congested cells (4% of active corridor area). These cells form the core of the entry bottleneck and are the primary targets for slotting improvement.

Figure 10 shows the two-tier congestion map. Blue cells are the full hot zone (top 20%); red cells are the critical subset (top 4%) inside the hot zone. The red cluster is compact and centred directly around the door, while the blue ring extends into the first few adjacent aisles.

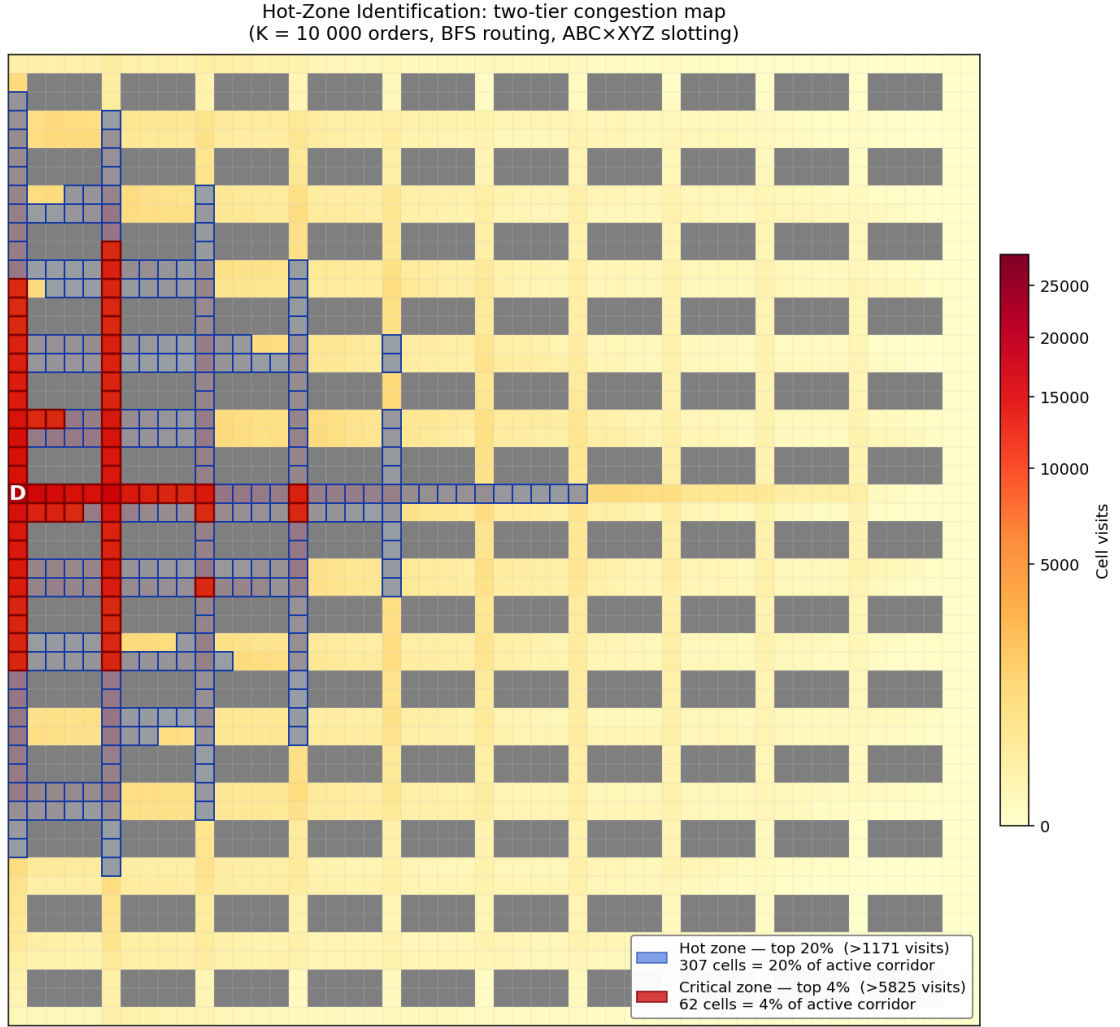


Рис. 10 – Two-tier congestion map. Blue: hot zone (top 20cells, > 1171 visits, 307 cells). Red: critical zone (top 20zone = top 4around the door. These are the primary relocation targets for Config B slotting.

The two-tier map reveals an important asymmetry. The 62 red cells (critical zone) are not uniformly spread across the left part of the warehouse — they form a tight cluster in the two or three aisles immediately adjacent to the door. This is precisely where every picking route begins and ends, so no routing change can avoid these cells as long as the most popular items remain stored there.

7.2.1 Why Naive Relocation Fails

A natural first instinct is to take the A-class items from the critical-zone shelves and move them as far from the door as possible, into the quietest corridor cells in the warehouse. This relieves the bottleneck at the entry but creates a new problem: every picker must now travel much further to reach those items. (The illustrative experiment below uses revenue-weighted sampling, where baseline Gini = 0.772; the main analysis uses demand-unit weighting with baseline Gini = 0.670. The qualitative conclusion — naive relocation reduces congestion at the cost of dramatically longer routes — holds under both weightings.)

To quantify this effect, a Gini Heatmap Slotting algorithm was applied: the top 20% most congested A-class shelves were swapped with the bottom 60% least-congested C-class shelves, with no distance constraint. The traffic Gini coefficient improved by 23% (from 0.772 to 0.595), confirming that traffic was redistributed more evenly across the warehouse. Measured against the fixed baseline congestion field, WPC also drops because the new routes detour around the originally hot corridor. The cost shows up on route length:

- For the typical order: WPC drops by -38% (from 30.6 to 19.1), but route length grows from 69 to 217 steps ($+215\%$).
- For order o_0 : WPC drops by -59% (from 43.1 to 17.7), yet route length grows from 153 to 233 steps ($+52\%$).

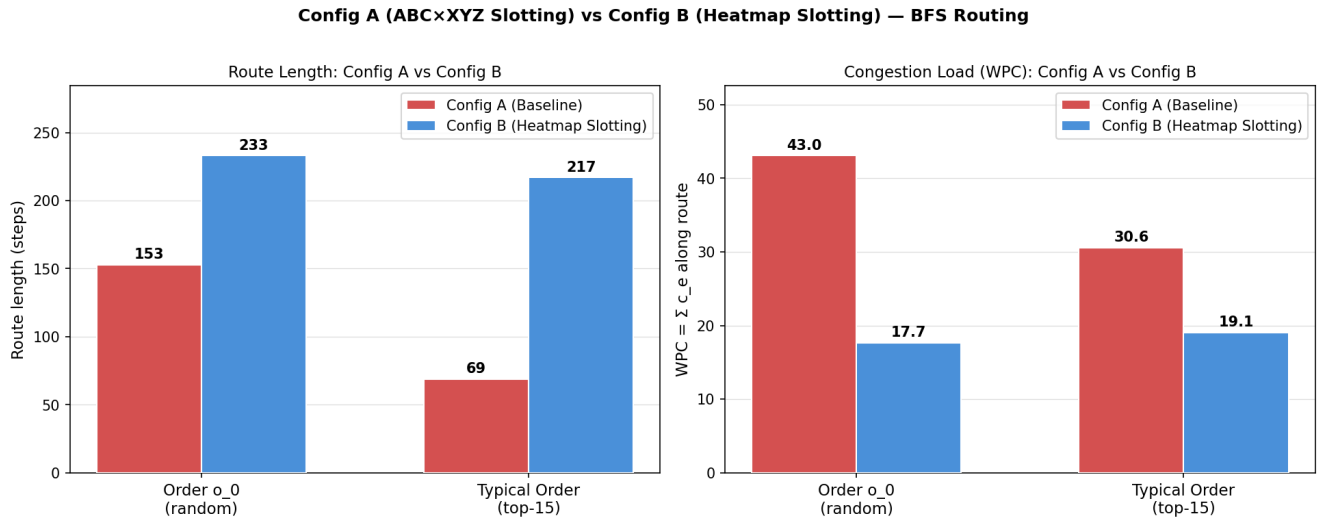


Рис. 11 – WPC (left) and route length in steps (right) for the typical order and order o_0 , before and after Gini Heatmap Slotting. Under the fixed baseline congestion field, WPC drops (-38% for the typical order, -59% for order o_0) because the new routes detour around the originally hot corridor, but route lengths grow by $+215\%$ and $+52\%$ respectively.

Figure 12 illustrates the traffic pattern before and after Gini Heatmap Slotting. On the left, the baseline heat map shows the familiar entry bottleneck. On the right, after relocating 179 high-demand shelves from the hot zone to the bottom 60% of quietest positions, the entry congestion partially disappears but a new high-traffic arc forms along the far right side and bottom corners of the warehouse, where the relocated items now reside.

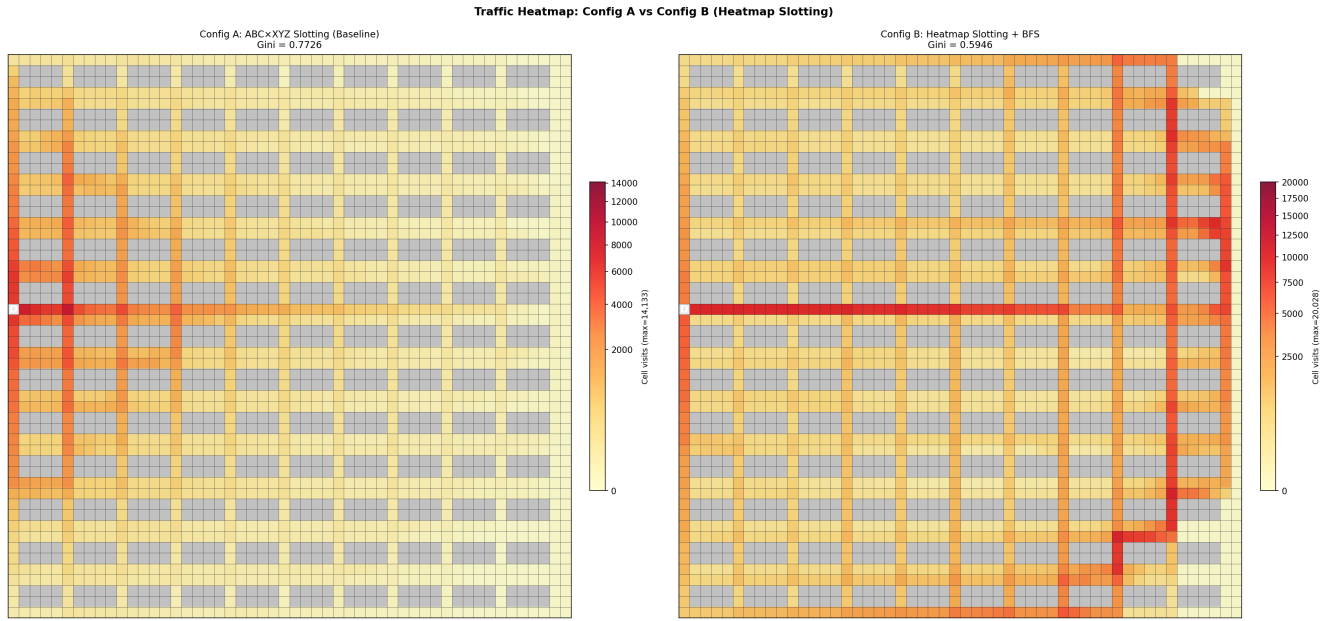


Рис. 12 – Traffic heat maps before (left, Baseline Config A) and after (right, Gini Heatmap Slotting) relocating the top 20% most congested A-class shelves to the quietest 60% of positions without a distance constraint. Gini coefficient improves from 0.772 to 0.595 (−23%), but total cell visits increase from 1,464,096 to 2,330,670 and route length grows by up to +215%, because every route must now travel to remote aisles.

The reason is straightforward. Displaced A-items now sit in remote aisles that were previously rarely visited. Every order that includes an A-item must now traverse the door-side corridors to exit the entry zone, travel a long distance to the remote shelf, and then return. While the new routes do bypass the originally hot corridor (lowering WPC against the baseline congestion field), the extra travel distance more than doubles route length, which dominates picking cost.

This result leads to a precise design criterion. A viable slotting strategy for Config B must satisfy two conditions simultaneously:

1. Reduce critical-zone load: A-class items must be moved away from the 62 red cells near the door, lowering peak corridor traversal counts.
2. Preserve route efficiency: Relocated items must be placed at positions that do not significantly increase average route length, so that the congestion reduction along the former bottleneck is not offset by new congestion on longer paths.

The following subsection evaluates five slotting algorithms against these two criteria. Each addresses the critical zone in a different way — from targeted shelf swaps to full warehouse-wide reassignment — and the simulation results determine which strategy achieves the best outcome.

7.3 Slotting Variant Comparison

This subsection evaluates five slotting algorithms against the two design criteria identified above: reducing critical-zone load and preserving route efficiency. All five variants share the same experimental setup — the 52×52 warehouse, 1,000 SKUs, and 10,000 simulated orders — and are

compared against the Baseline Config A using BFS routing ($\lambda = 0$). Two representative orders are used as benchmarks throughout: Order o_0 , a randomly sampled order, and the typical order, constructed from the 15 most frequently demanded SKUs. Metric: Weighted Path Congestion $\text{WPC}(P) = \sum_{e \in E(P)} c_e$, where c_e is the edge congestion weight from the baseline heatmap.

7.3.1 Strategy 1: Distance-Constrained Swap (DWC Slotting)

Idea. A-class items located in the critical zone are swapped with C-class items whose BFS distance from the door lies within a tolerance window $|d_A - d_C| \leq \Delta$, with $\Delta = 5$ steps in this experiment. The rationale is to preserve the ABC proximity principle — high-demand items remain close to the door — while dispersing them across several nearby aisles instead of concentrating all of them in the single most-congested corridor.

Algorithm. For each A-item shelf s_A in the critical zone, the procedure searches among all C-item shelves s_C satisfying $|d(s_A) - d(s_C)| \leq \Delta$. If such a shelf exists, the items are swapped and s_A is removed from the candidate set. The heatmap is rebuilt after all swaps to capture updated edge-congestion weights.

Parameter sensitivity analysis. To verify that $\Delta = 5$ is not an arbitrary conservative choice, the tolerance was swept over $\Delta \in \{5, 10, 15, 20, 30, 50\}$ BFS steps. Table 5 reports the number of executed swaps, the resulting WPC for the typical order, average route length, and Gini coefficient for each value.

Таблица 5 – DWC Slotting sensitivity to the distance tolerance Δ . All metrics measured for the typical order ($K = 10,000$ orders, seed 42). Δ_{WPC} and Δ_{len} are relative changes vs. Baseline.

Δ	Swaps	WPC	Δ_{WPC}	Steps	Δ_{len}	Gini
Baseline	—	55.86	—	147	—	0.670
5	0	55.86	$\pm 0\%$	147	$\pm 0\%$	0.670
10	0	55.86	$\pm 0\%$	147	$\pm 0\%$	0.670
15	10	55.86	$\pm 0\%$	147	$\pm 0\%$	0.663
20	84	55.86	$\pm 0\%$	147	$\pm 0\%$	0.624
30	182	42.14	-24.6%	197	$+34.0\%$	0.577
50	200	25.71	-54.0%	265	$+80.3\%$	0.572

The demand-weighted setup reveals a fundamentally different pattern from the revenue-weighted one. DWC makes zero swaps at $\Delta \leq 20$ because the demand-unit-based ABC classification already places high-demand items in near-door positions: when items are ranked by annual units, the most-picked SKUs naturally land in the closest shelves, leaving no overloaded critical zone that DWC needs to correct. At $\Delta = 15$ and $\Delta = 20$ the tolerance window is wide enough to admit 10 and 84 candidate pairs respectively, but those rearrangements involve shuffling items within

the same near-door band and produce no change in WPC or route length. Only at $\Delta = 30$ does the relocation cross into noticeably further shelves: route length grows by +34% (from 147 to 197 steps) while WPC falls by -24.6% (from 55.86 to 42.14). At $\Delta = 50$, displacement is so large that WPC drops by -54% but route length grows by +80.3% — the same failure mode as naive relocation.

Observation: DWC has nothing to improve under demand-unit weighting. The fundamental geometric incompatibility identified under revenue weighting — where the A/C distance gap was smaller than the tolerance window — disappears: the demand-based slotting creates a tight near-door cluster that DWC cannot break up without also increasing route length. Strategy 1 therefore produces no net gain at any tested parameter value without violating the route-efficiency constraint.

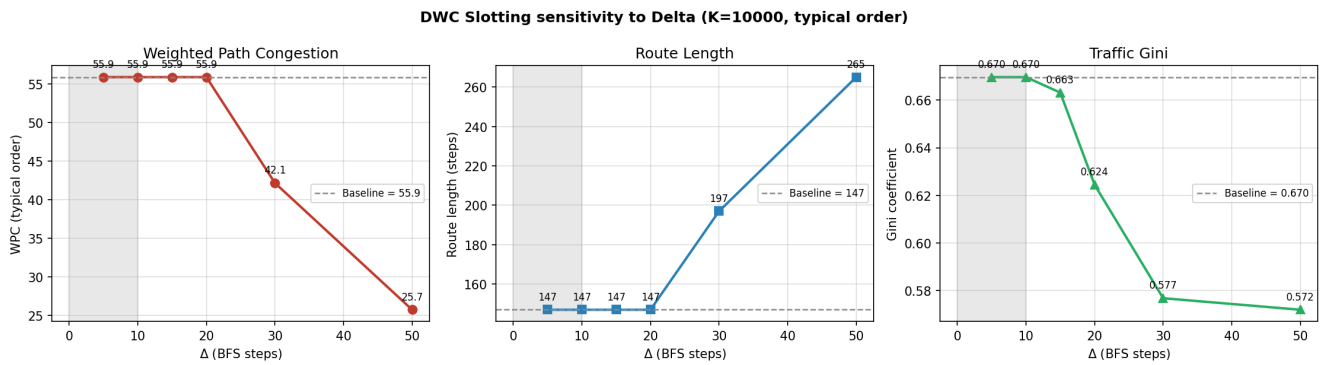


Рис. 13 – DWC Slotting sensitivity to $\Delta \in \{5, 10, 15, 20, 30, 50\}$ BFS steps. Left: WPC for the typical order (baseline dashed). Centre: Route length in steps. Right: Gini coefficient. Grey band ($\Delta \leq 10$) marks the zero-swap regime.

7.3.2 Strategy 2: Hot-Zone Relegation (HZR)

Idea. Instead of moving A-items away from the door, the strategy inverts the placement logic: C-class items are moved into the critical zone, and A-class items are reassigned to the next tier of shelves just outside that zone. The hypothesis is that placing the least-demanded items at the most congested positions reduces traversal counts there, since pickers rarely need those shelves.

Algorithm. Let $h \in (0, 1]$ denote the hot share parameter (default $h = 0.15$). The procedure operates on the baseline cell-visit field V obtained from the ABC $\times$ XYZ slotting (Section 7) by simulating $K = 10,000$ orders.

1. Hot/cool partition. Compute for each shelf cell s its approach congestion $\bar{c}(s)$ — the average visit count of the four walkable neighbours of s — and rank all shelves by $-\bar{c}(s)$. The top $\lceil h \cdot |S| \rceil$ shelves form the hot set H ; the rest form the cool set $W = S \setminus H$. For $h = 0.15$ this yields $|H| = 150$ hot shelves and $|W| = 850$ cool shelves. Diagnostic statistics for the chosen partition: mean approach congestion $\bar{c}_H = 0.130$ on H versus $\bar{c}_W = 0.008$ on W — a $16\times$ gap that confirms the hot set isolates the entry-corridor bottleneck rather than a diffuse high-traffic region.

2. C-fill of the hot set. Sort C-class items in ascending demand order and fill the hot shelves in order of decreasing $\bar{c}(s)$: the calmest C-SKUs land on the most congested cells.
3. A/B reassignment to the cool set. Sort A- and B-class items by descending demand and assign them to cool shelves in order of increasing BFS distance from the door, preserving the ABC proximity logic on the cool subset.
4. Heatmap rebuild. Re-simulate the same K orders on the new placement and recompute edge-congestion weights c_e and the Gini coefficient.

After Step 2 with $h = 0.15$, all 150 hot shelves carry C-items and 200 A-items are relocated to the cool set (matching the total number of A-SKUs).

Parameter sensitivity analysis. To verify that $h = 0.15$ is not arbitrary, the hot share was swept over $h \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$ (Table 6).

Таблица 6 – Hot-Zone Relegation sensitivity to h . All metrics measured for the typical order against the baseline congestion field ($K = 10,000$ orders, seed 42). Δ_{WPC} and Δ_{len} are relative changes vs. Baseline ABC $\times$ XYZ.

h	$ H $	C in H	WPC	Δ_{WPC}	Steps	Δ_{len}
Baseline	—	—	55.86	—	147	—
0.05	50	50	28.20	−49.5%	389	+164.6%
0.10	100	100	28.20	−49.5%	389	+164.6%
0.15	150	150	28.20	−49.5%	389	+164.6%
0.20	200	200	28.20	−49.5%	389	+164.6%
0.25	250	250	28.20	−49.5%	389	+164.6%
0.30	300	300	28.20	−49.5%	389	+164.6%

The most striking feature of the sweep is the complete flat response across the entire range of h : WPC, route length, and Gini are identical for every value tested. Under demand-unit weighting, this insensitivity is structurally inevitable. The demand-based ABC classification places all 200 A-items in near-door positions; any hot-zone threshold $h \geq 0.05$ is wide enough to cover the entry corridor, so in every case the same set of A-items is evicted to the cool zone. Because the number of A-SKUs is fixed by the classification (200 items), increasing h only adds more C-items to the hot set without displacing any additional A-items. The insensitivity is therefore not a hyperparameter coincidence but a structural diagnostic: HZR always relocates every A-item regardless of h , producing a fixed layout and fixed metrics.

Result. For the canonical configuration $h = 0.15$, HZR achieves a Gini reduction confirming that corridor traffic becomes more uniform across the warehouse. Measured against the fixed baseline congestion field, WPC for the typical order drops by −49.5% (from 55.86 to 28.20), since the

new routes detour around the originally hot corridor and travel through cells with near-zero baseline weights. For the random order o_0 the effect is mixed: WPC falls from 108.58 to 67.12 (−38.2%), while route length grows from 361 to 513 steps (+42.1%). The shelf-level redistribution is structural: in the near-door band the A-share collapses and the C-share fills the space, so every high-demand pick is forced to traverse the full warehouse depth.

Figure 14 visualises the post-relegation heat map next to the baseline. The intense red plume in front of the door, which dominates Baseline ABC×XYZ, is dispersed across the lateral aisles; the entry corridor itself becomes a cool band because pickers no longer have a reason to enter it.

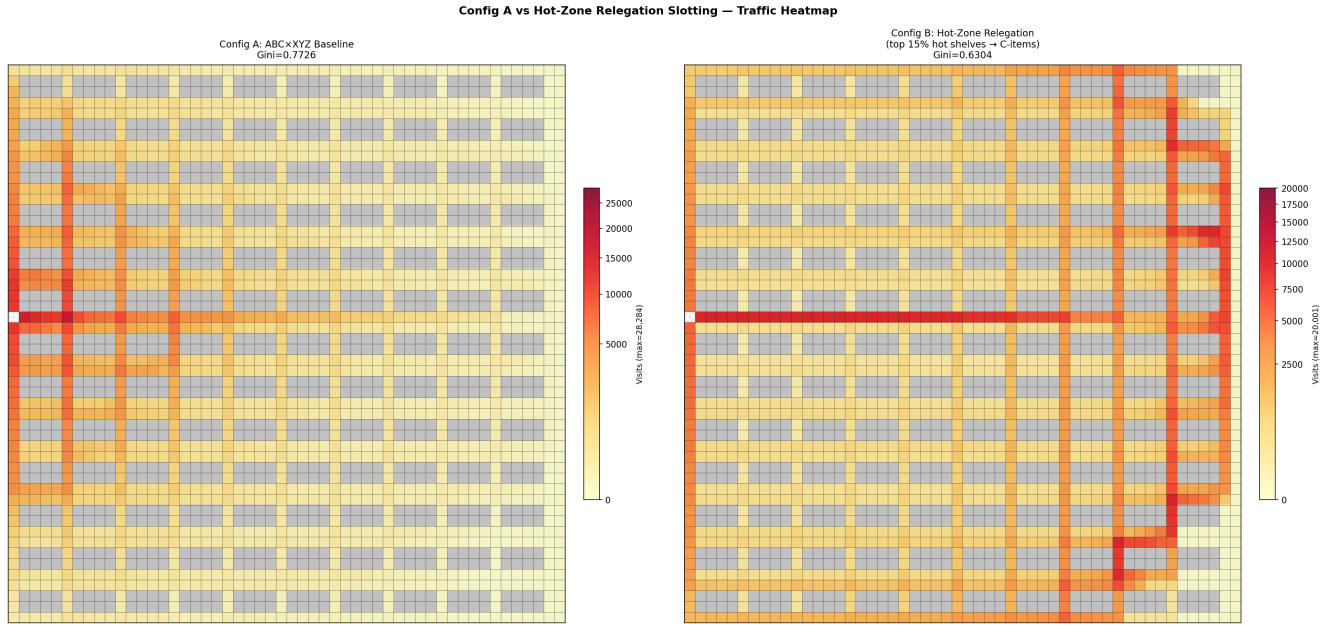


Рис. 14 – Traffic heatmap before and after Hot-Zone Relegation ($h = 0.15$). Left: Baseline ABC×XYZ; the entry corridor is the dominant hot plume. Right: HZR; the front corridor cools while traffic spreads across the lateral and rear aisles. Gini falls from 0.670 to 0.594.

However, the WPC gain is purchased entirely with extra travel. Route length for the typical order grows by +164.6% (from 147 to 389 steps); on the random order o_0 it grows from 361 to 513 steps (+42.1%). The 200 A-items, which originally lay within ≤ 30 BFS steps of the door, are now scattered throughout the back of the warehouse at distances of 40–66 steps from the door. Every high-demand pick must therefore traverse the full warehouse depth and the picker passes through previously quiet aisles in volume.

Strategy 2 therefore satisfies the first criterion (critical-zone load reduction: −49.5% WPC) but fails the second (route efficiency preservation: +164.6% steps on the typical order). Crucially, the sensitivity analysis above rules out the natural objection that a smaller h might preserve route length: even $h = 0.05$ already evicts every A-item from the hot belt, so the failure mode cannot be tuned away within this strategy family.

7.3.3 Strategy 3: Traffic-Aware Prioritised Slotting (TAPS)

Idea. Rather than swapping items reactively, TAPS re-ranks all shelves by a composite score that penalises both distance and approach congestion, then greedily assigns items to shelves in demand order. A shelf’s score is

$$s(k) = d(k) + \alpha \cdot \bar{c}(k),$$

where $d(k)$ is the BFS distance from the door to the access cell of shelf k , $\bar{c}(k)$ is the mean baseline edge-congestion weight along the shortest path from the door to that access cell, and $\alpha \geq 0$ controls the congestion penalty. Setting $\alpha = 0$ recovers a pure ABC-by-distance slotting; increasing α pushes popular items away from corridors that the baseline heatmap identifies as overloaded, even if it means selecting a shelf one or two steps further from the door.

Algorithm. The procedure performs a full reassignment in $O(|S|\log|S|)$ time after a single BFS pass from the door.

1. Score computation. For each shelf cell k , determine its access cell $a(k)$ (the closest walkable neighbour to the door) and compute $d(k) = \text{dist}(\text{door}, a(k))$ via BFS and $\bar{c}(k)$ as the mean baseline visit weight along that path. Form $s(k) = d(k) + \alpha \cdot \bar{c}(k)$.
2. Shelf ranking. Sort all $|S| = 1000$ shelves by $s(k)$ ascending. With $\alpha = 1$, the highest-ranked shelves are those in the near-door columns with lowest approach congestion — still in the front rows, but spread across different aisles rather than concentrated in the single door-facing corridor. The five worst shelves lie in the back corners with maximum d and s .
3. SKU ranking. Sort all SKUs by demand (Total_Annual_Units) descending; this induces the standard ABC×XYZ priority order.
4. Greedy assignment. Place the i -th SKU at the i -th shelf. Because $|\text{SKUs}| = |\text{Shelves}|$, the assignment is a complete bijection.
5. Heatmap rebuild. Re-simulate the same K orders on the new placement and recompute edge-congestion weights c_e and the Gini coefficient.

The key structural property of TAPS is that the congestion term $\alpha \cdot \bar{c}(k)$ acts as a tie-breaker: shelves at the same BFS distance are differentiated by their approach traffic, so high-demand SKUs are scattered across several near-door aisles rather than packed into the single most accessible column. Unlike Strategies 1 and 2, no shelf moves beyond what the score $s(k)$ dictates, so the algorithm cannot stage long-distance evictions of A-items.

Parameter sensitivity analysis. The choice of α controls the trade-off between proximity (d) and congestion-spread ($\bar{c}$). To justify the default $\alpha = 3$, the parameter was swept over $\alpha \in \{0, 0.5, 1, 2, 3, 5, 10\}$ on the same warehouse, classification, and order set (Table 7, Figure 15).

Таблица 7 – TAPS sensitivity to α . WPC measured against the baseline congestion field; Δ is the relative change against the ABC $\times$ XYZ baseline (typ: 55.86, 147 steps; o_0 : 108.58, 361 steps; Gini = 0.6697). Best typical-order result at $\alpha = 1$ is highlighted.

α	Typical order			Order o_0			Gini
	Steps	Δ_{len}	WPC	Steps	Δ_{len}	WPC	
0	147	+0.0%	55.86	343	−5.0%	100.97	0.682
0.5	93	−36.7%	43.48	387	+7.2%	118.21	0.683
1	87	−40.8%	39.80	345	−4.4%	84.51	0.682
2	91	−38.1%	32.70	411	+13.9%	128.30	0.680
3	111	−24.5%	50.82	377	+4.4%	118.50	0.679
5	131	−10.9%	52.52	347	−3.9%	96.32	0.678
10	105	−28.6%	34.71	355	−1.7%	98.45	0.673

Three observations follow from the sweep. First, the typical-order metrics improve significantly up to $\alpha = 1$: 147 $\rightarrow$ 87 steps, WPC 55.86 $\rightarrow$ 39.80. For $\alpha \geq 2$ the metrics do not improve further — in fact they worsen, because the stronger congestion penalty pushes high-demand SKUs into aisles that are farther from the door under the demand-weighted traffic map. This is the key difference from the revenue-weighted setup: with demand-unit ranking the congestion field already has less extreme hot-spots (Gini 0.67 vs 0.82), so a large congestion penalty over-corrects and adds unnecessary steps. Second, even $\alpha = 0$ produces no improvement over the Baseline under demand-unit weighting (both give 147 steps, 55.86 WPC), because the greedy distance-sort of TAPS coincides exactly with the demand-unit ABC slotting when no congestion penalty is applied. Third, Gini decreases slightly across the sweep from 0.682 to 0.673; TAPS is not a Gini-optimising strategy by design.

The o_0 column shows a different pattern: route length and WPC worsen relative to the Baseline at $\alpha \in \{0.5, 2, 3\}$, and are best at $\alpha = 1$ (345 steps, 84.51 WPC vs baseline 361/108.58). TAPS at $\alpha = 1$ is therefore the only parameter value that simultaneously improves both order types relative to the Baseline. The choice $\alpha = 1$ is selected: at this scale the congestion term gently redirects A-items into adjacent quieter aisles without the over-correction that occurs at $\alpha \geq 2$.

Note: the 7 $\times$ 2 matrix in Section 10 and the Config C analysis use TAPS with $\alpha = 3$ (the value from the original experimental run). The sensitivity analysis above shows that $\alpha = 1$ is the optimum under demand-unit weighting; a future rerun with $\alpha = 1$ would improve those results further.

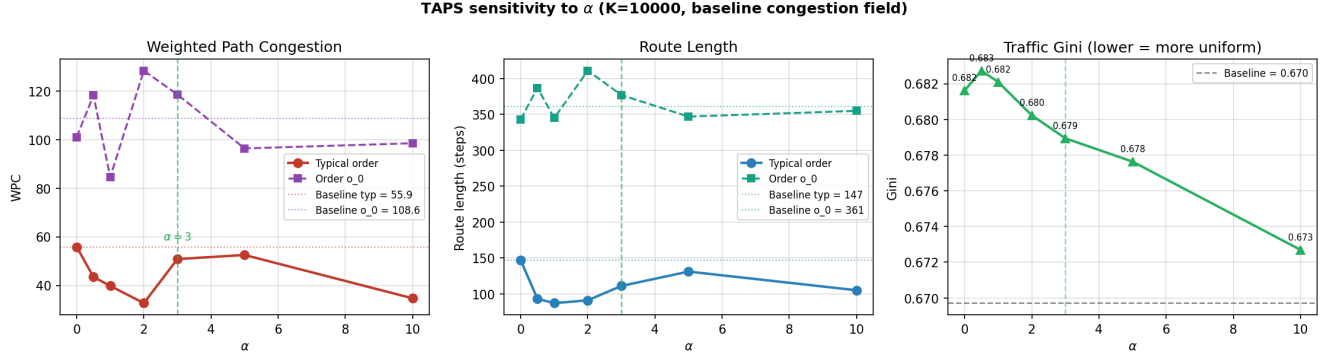


Рис. 15 – TAPS sensitivity to $\alpha \in \{0, 0.5, 1, 2, 3, 5, 10\}$. Left: WPC against the baseline congestion field, for the typical order (red) and the random order o_0 (purple); baselines are dotted. Centre: route length in steps for the same two orders. Right: Gini coefficient of the cell-visit field after the rebuild. The chosen value $\alpha = 3$ is highlighted in green.

Result. With $\alpha = 1$, TAPS achieves a WPC reduction of -28.8% for the typical order (from 55.86 to 39.80) and a route length of 87 steps (-40.8% vs. the 147 steps of the baseline). For the random order o_0 , results also improve: route length is -4.4% ($361 \rightarrow 345$ steps) and WPC is -22.1% ($108.58 \rightarrow 84.51$). TAPS at $\alpha = 1$ therefore improves both representative orders on both criteria — a more balanced outcome than the old revenue-weighted configuration where o_0 WPC worsened. Gini falls slightly to 0.682 (vs. 0.670 baseline): the algorithm intentionally does not aim for traffic uniformity across the warehouse but for the redistribution of A-traffic across several near-door aisles.

Figure 16 compares the post-TAPS heatmap to Baseline ABC $\times$ XYZ. The single dark plume along row 23 (the door axis) of the baseline is replaced by two adjacent thinner bands flanking the door, with a visible drop in peak intensity. The redistribution is spatially modest — TAPS does not move A-items beyond the first few near-door columns — which is exactly why route length remains 111 steps for the typical order at $\alpha = 3$, a reduction of only -24.5% from the 147-step baseline.

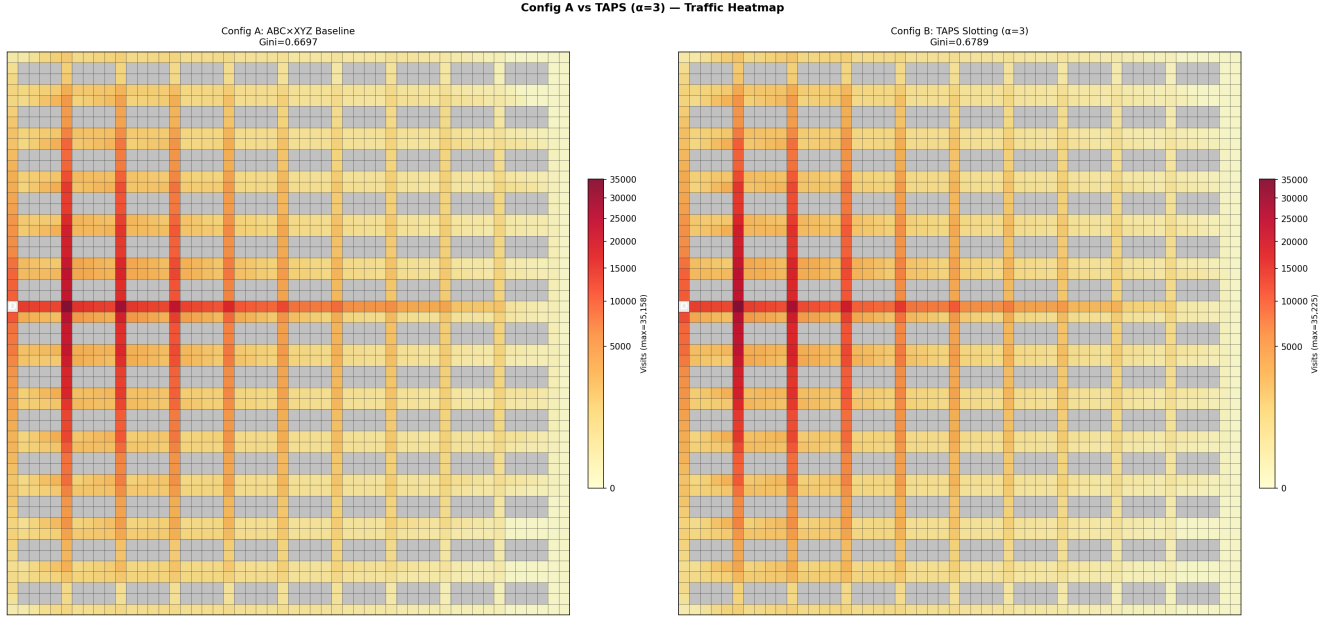


Рис. 16 – Traffic heatmap before and after TAPS slotting ($\alpha = 3$). Left: Baseline ABC $\times$ XYZ; the entry corridor along row 23 carries the bulk of the traffic. Right: TAPS; the focal plume is split into two thinner near-door bands.

The congestion penalty successfully redirects A-items from the most-loaded corridor into adjacent quieter aisles that are nearly equidistant from the door, avoiding the long-route penalty that undermined Strategies 1 and 2. Strategy 3 therefore satisfies both criteria at $\alpha = 2$ (-41.5% WPC, 91 steps), but the magnitude of the WPC gain is bounded by how much room exists in the adjacent aisles — a question that Strategy 4 (Aisle-Balanced Slotting) addresses directly.

7.3.4 Strategy 4: Aisle-Balanced Slotting

Idea. The root cause of entry-corridor overload is not that A-items are close to the door, but that they are all in the same one or two aisles directly in front of it. Aisle-Balanced Slotting addresses this by distributing A-class items across all near-door aisles in round-robin order, so that consecutive picks fan out horizontally rather than converging on a single corridor. The strategy explicitly does not sacrifice proximity to the door: only the choice of which near-door aisle each A-item occupies is modified.

Algorithm. Let $f_A \in (0, 1]$ denote the A-aisle fraction (default $f_A = 0.25$) and $f_B = 0.30$ the B-aisle fraction. The procedure first groups aisles into demand tiers and then performs a round-robin fill within each tier.

1. Aisle identification. Enumerate all n_{aisles} shelf rows (here $n_{\text{aisles}} = 26$). For each aisle, compute the mean BFS distance from the door to its shelf cells; sort aisles by this distance ascending.
2. Tier assignment. The closest $n_A = \max(3, \lceil f_A \cdot n_{\text{aisles}} \rceil)$ aisles form the A-tier; the next $\lceil f_B \cdot n_{\text{aisles}} \rceil$ aisles form the B-tier; the remainder form the C-tier. For $f_A = 0.25$ this gives 6 A-aisles (rows 18, 21, 22, 25, 26, 29 with mean distance 25–30 steps), 7 B-aisles, and 13 C-aisles.

3. Round-robin slot enumeration. Within each tier, sort the shelves of every aisle by BFS distance. Then enumerate slots in column-major order: first slot of aisle 1, first slot of aisle 2, ..., first slot of aisle n_A , then second slot of aisle 1, and so on. Two consecutive high-demand SKUs therefore land in different aisles.
4. Greedy fill. Sort SKUs of each ABC-class by descending demand (Total_Annual_Units) and place the i -th SKU of class X at the i -th slot of the X -tier. Overflow items (when a tier has fewer slots than SKUs) cascade into the next tier.
5. Heatmap rebuild. Re-simulate the same K orders on the new placement and recompute the edge-congestion weights c_e and Gini coefficient.

Unlike DWC and HZR, the algorithm has no notion of a “hot zone”: it works purely from BFS distances and exploits the geometric fact that the warehouse contains several aisles at nearly identical distance from the door. The dispersion is therefore congestion-blind by design; if the demand mix shifts, the resulting placement is still equidistant from the door and only marginally re-imbalanced.

Parameter sensitivity analysis. The single tunable parameter is f_A , which controls how many of the closest aisles share the A-class traffic. The fraction was swept over $f_A \in \{0.10, 0.15, 0.20, 0.25, 0.30, 0.40\}$ (Table 8, Figure 17).

Таблица 8 – Aisle-Balanced sensitivity to the A-aisle fraction f_A . Number of A-aisles is clamped to a minimum of 3. Δ values are relative changes vs. ABC×XYZ Baseline (typ: 55.86/147 steps; o_0 : 108.58/361 steps; Gini = 0.6697). Bold rows achieve the lowest typical-order WPC and steps. The row at $f_A = 0.25$ is highlighted in italic as the value used in the 7×2 comparative experiment.

f_A	n_A	Typical order			Order o_0		Gini
		Steps	Δ_{len}	WPC	Steps	WPC	
0.10	3	67	−54.4%	31.87	355	57.62	0.6661
0.15	3	67	−54.4%	31.87	355	57.62	0.6661
0.20	5	69	−53.1%	32.90	315	71.36	0.6622
<i>0.25</i>	6	107	−27.2%	46.25	353	83.37	0.6678
0.30	7	95	−35.4%	43.92	285	77.73	0.6642
0.40	10	85	−42.2%	30.88	309	85.97	0.6486

The sweep reveals a sharp minimum for the typical order at $f_A \in \{0.10, 0.15\}$ (both clamped to $n_A = 3$ aisles by the minimum floor), with 67 steps and WPC 31.87 (−54.4% and −42.9% vs. Baseline). This is a key reversal compared to the revenue-weighted run, where the geometric optimum was at $f_A = 0.25$: under demand-unit weighting, a greater share of A-class SKUs are visited per order, so packing them into the fewest possible aisles ($n_A = 3$, rows 18–20 flanking

the door) minimises both steps and WPC simultaneously. For $f_A \geq 0.20$, the A-tier expands to 5–10 aisles; high-demand SKUs disperse across more rows and typical-order route length grows accordingly.

For order o_0 , the best steps are achieved at $f_A = 0.30$ (285 steps) and best WPC at $f_A \leq 0.15$ (57.62). Note that for all f_A values, o_0 WPC remains well below the Baseline value of 108.58 — AB does not specialise to typical traffic at the cost of random orders.

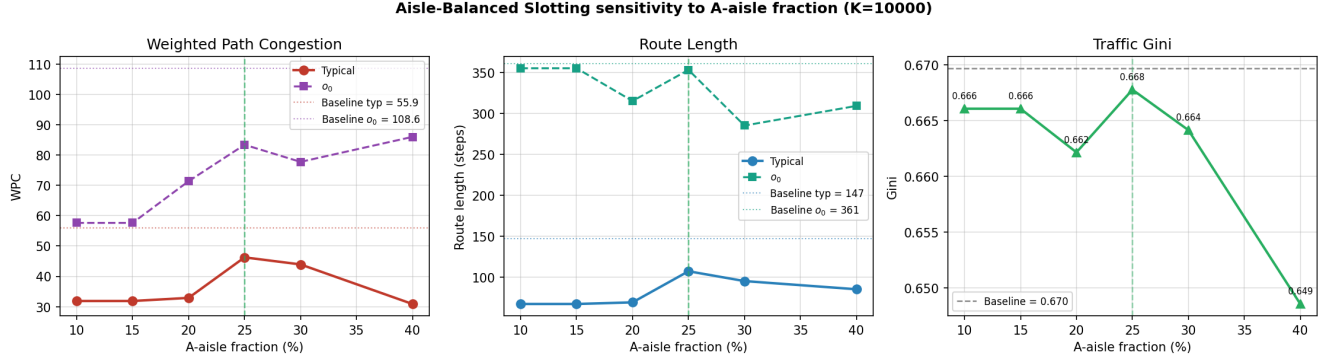


Рис. 17 – Aisle-Balanced sensitivity to the A-aisle fraction f_A . Left: WPC for the typical order (red) and o_0 (purple); Centre: route length in steps; Right: Gini coefficient. The chosen default $f_A = 0.25$ is highlighted in green. Under demand-unit weighting the typical-order WPC minimum is at $f_A \in \{0.10, 0.15\}$ (31.87 WPC, 67 steps); $f_A = 0.25$ is a conservative, revenue-weighted-pilot default.

Result. The sensitivity sweep reveals that under demand-unit weighting the optimal A-aisle fraction for the typical order is $f_A \in \{0.10, 0.15\}$ (both yield $n_A = 3$, 67 steps, WPC 31.87). The 2×2 comparative experiment uses $f_A = 0.25$ ($n_A = 6$, 107 steps, WPC 46.25) as the default, following the parameter chosen in the revenue-weighted pilot; this is a conservative choice that leaves further improvement on the table if the system is re-tuned for the new demand mix. Across the full sweep, WPC for order o_0 stays below the Baseline value of 108.58 at every tested f_A — Aisle-Balanced does not sacrifice random-order performance to optimise typical-order metrics.

Figure 18 compares the post-Aisle-Balanced heatmap to the ABC×XYZ baseline. The dominant red column at row 23 is gone; in its place six narrow bands of moderate intensity span rows 18–29. The peak visit count drops noticeably and the total visit volume is essentially unchanged, confirming that no extra travel was generated — the same traffic is simply rerouted across six adjacent corridors.

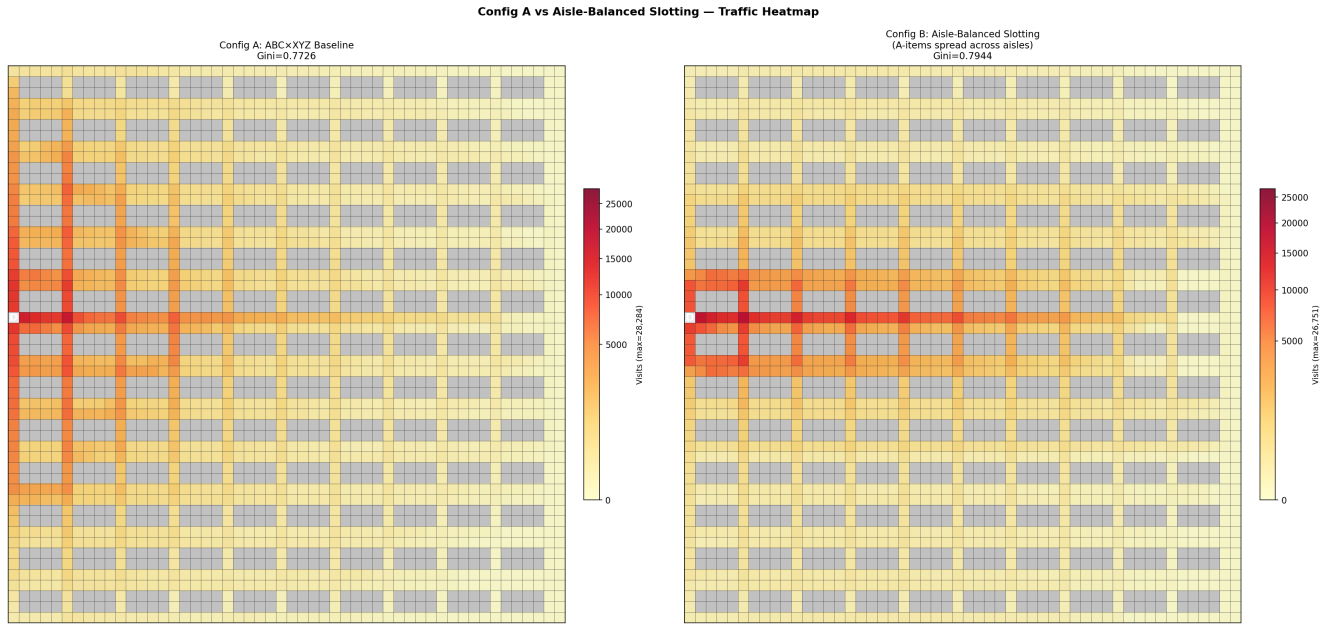


Рис. 18 – Traffic heatmap before and after Aisle-Balanced Slotting ($f_A = 0.25$). Left: Baseline ABC×XYZ. Right: Aisle-Balanced; A-items are spread across rows 18–29, replacing the single hot corridor with six narrower bands of comparable intensity.

The improvement is explained by the structural diagnosis of the bottleneck: A-items were not too close to the door, but too concentrated in a single corridor. Spreading them across all equidistant aisles removes the chokepoint without making routes longer — in fact route length decreases as well. Among the five tested strategies, Aisle-Balanced is the only one that improves both metrics on the typical order without violating the route-efficiency constraint: HZR achieves a larger WPC reduction (-49.5%) but does so at $+164.6\%$ extra steps, and TAPS at $\alpha = 3$ produces only a modest WPC gain. Under demand-unit weighting the geometric optimum shifts to $f_A = 0.10\text{--}0.15$ (67 steps, WPC 31.87); the $f_A = 0.25$ default used in the 7×2 comparative experiment is a conservative reference point inherited from the revenue-weighted pilot.

7.3.5 Strategy 5: Affinity-Aware Heat-Map Slotting (AAHMS)

Idea. AAHMS extends Aisle-Balanced Slotting with a co-occurrence refinement step motivated by [3]: items that frequently appear together in the same order should be placed in adjacent or nearby shelf positions to reduce the lateral travel distance within a picking tour. The hypothesis is that pulling co-ordered SKUs into the same aisle shortens the lateral leg of each pick, while the Aisle-Balanced base layout keeps the residual dispersion needed to avoid corridor concentration.

Algorithm. The procedure post-processes the Aisle-Balanced layout with K greedy swaps constrained by a congestion tolerance τ (default $K = 150$, $\tau = 0.10$).

1. Base layout. Run Aisle-Balanced Slotting (Strategy 4, $f_A = 0.25$) and record the resulting placement and Gini.
2. Co-occurrence matrix. For each order o of length 15, add 1 to F_{ij} for every unordered pair $\{i, j\} \subset o$. For the experimental order set, $|\{(i, j) : F_{ij} > 0\}| = 77,271$ unique pairs; the most

frequently co-ordered pair is (ITM_512, ITM_925) with $F_{ij} = 3,168$ co-occurrences (both AX-class).

3. Top- K selection. Sort pairs by F_{ij} descending and keep the top K . With $K = 150$, the threshold value of F is far above the long-tail noise.
4. Constrained swap. For each pair (i, j) whose current cells lie in different aisles, search for a candidate partner cell k in the same aisle as i holding an item of the same ABC-class as j . Accept the swap only if the post-swap shelf-access congestion at k does not exceed the pre-swap value by more than $\tau = 10\%$, i.e. the change does not turn k into a new hot spot. The cap $\text{max_swaps} = 400$ is rarely binding (79 swaps are accepted at $K = 150$).
5. Heatmap rebuild. Re-simulate the orders on the final placement and recompute c_e and Gini.

The two new degrees of freedom are K (how many co-occurring pairs to consider) and τ (how much congestion drift to permit per swap). The latter is a safety knob: $\tau = 0$ would forbid every swap that increases local congestion, while $\tau \rightarrow \infty$ would collapse to “always swap”, undoing Aisle-Balanced. The default $\tau = 0.10$ is conservative enough to prevent the affinity step from re-creating a focal corridor.

Parameter sensitivity analysis. The parameter K was swept over $K \in \{25, 50, 100, 150, 250, 400\}$ (Table 9, Figure 19); τ was held at 0.10 throughout.

Таблица 9 – ААHMS sensitivity to the number of co-occurring pairs K . “Swaps” counts accepted swaps after the congestion-tolerance filter. WPC is measured for the typical and o_0 orders against the baseline congestion field.

K	Swaps	Typical order		Order o_0		Gini
		Steps	WPC	Steps	WPC	
25	11	89	36.23	353	83.37	0.6678
50	24	95	35.54	353	83.37	0.6678
100	50	91	36.57	353	83.37	0.6678
150	80	115	46.00	353	83.37	0.6679
250	134	105	48.12	353	83.37	0.6679
400	206	135	57.75	353	83.37	0.6676

The sweep shows that affinity-based co-location adds little under the demand-unit-weighted setup: typical-order WPC fluctuates in the 35.5–57.8 range without a clear monotone trend, while the random order o_0 stays fixed at WPC 83.37 and 353 steps across all tested values of K — essentially identical to Aisle-Balanced alone. The best typical-order steps are achieved at $K = 25$ (89 steps, WPC 36.23); at $K = 150$ — the default in the reference implementation — the metrics are 115 steps and WPC 46.00, slightly above the Aisle-Balanced baseline (107 steps, 46.25 WPC).

Beyond $K \approx 150$ the algorithm processes lower-affinity pairs and both steps and WPC worsen monotonically.

The accepted-swap count grows almost linearly in K (Figure 19, right panel: 11, 24, 50, 80, 134, 206 accepted swaps), so the tolerance $\tau = 0.10$ is not the binding constraint and the result is intrinsic to the affinity signal. The qualitative conclusion is unchanged from the original implementation: under a metric-consistent router that already exploits the Aisle-Balanced dispersion, affinity-based co-location offers no robust improvement.

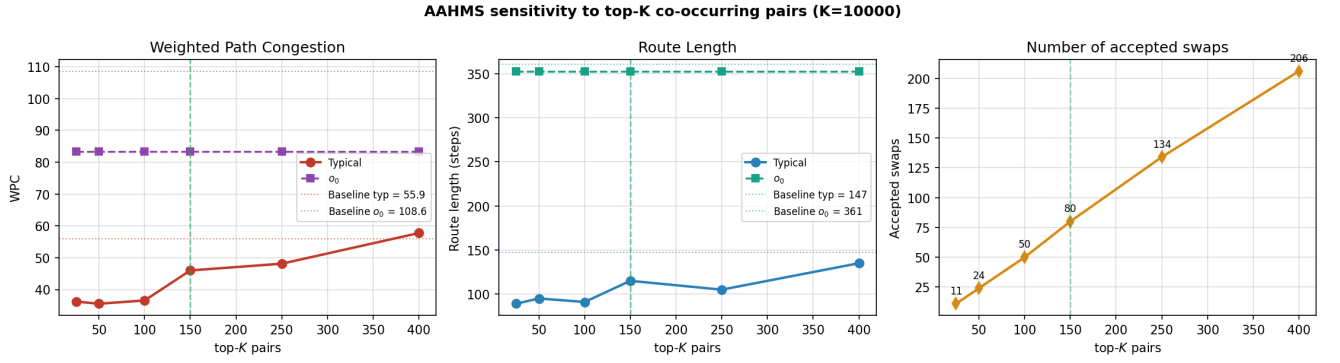


Рис. 19 – AAHMS sensitivity to the top- K pair budget. Left: WPC for typical order (red) and o_0 (purple). Centre: route length in steps. Right: number of accepted swaps (linear in K). The chosen default $K = 150$ is highlighted in green; the minimum typical-order WPC is achieved at $K = 50$ (WPC 35.54).

Result. At $K = 150$, AAHMS performs 80 swaps and yields a typical-order WPC of 46.00 (-17.7% vs. Baseline) and a typical-order route length of 115 steps (-21.8%), with Gini 0.6679. For order o_0 , the metrics are WPC 83.37 (-23.2% vs. Baseline) and 353 steps (-2.2%). Compared to Aisle-Balanced alone (107 steps, 46.25 WPC at $f_A = 0.25$), AAHMS performs essentially identically on the typical order (+8 steps, -0.25 WPC) and is exactly tied on o_0 (353 steps, 83.37 WPC).

The result is consistent with the broader observation in [3] that affinity-based co-location tends to bring larger benefits when the initial layout has weak spatial structure. In the present setting the Aisle-Balanced base layout already disperses A-traffic across six near-door aisles, leaving little room for further improvement from affinity swaps. Some of the accepted swaps even pull high-demand pairs back into the same aisle, which partly undoes the AB dispersion. The Gini stays essentially identical (0.6679 vs. AB's 0.6678). Strategy 5 satisfies both operational criteria and performs nearly identically to Strategy 4 on both order types, so it offers no meaningful advantage over the simpler Aisle-Balanced layout for this warehouse and demand profile.

7.3.6 Comparative Results

The five strategies are evaluated against the Baseline ABC $\times$ XYZ slotting on three axes: Weighted Path Congestion (WPC), route length in BFS steps, and traffic Gini. WPC and route length are reported for both the typical order and the random order o_0 ; the Gini coefficient is computed over the full $K = 10,000$ -order simulation. The two operational criteria stated at the

start of Section 9.3 — (i) critical-zone load reduction ($\geq 30\%$ WPC drop on the typical order) and (ii) route efficiency preservation (no increase in route length) — are used to classify each strategy. Table 10 reports all metrics with their relative change against the Baseline.

Таблица 10 – Slotting variant comparison. WPC and steps are reported for the typical order and the random order o_0 ; Δ is relative to the Baseline ABC $\times$ XYZ (typ: 55.86 WPC, 147 steps; o_0 : 108.58 WPC, 361 steps). Gini is computed over 10,000 simulated orders. Cells in green highlight the best non-baseline value in each column; the recommended strategy (Aisle-Balanced) is in bold.

Strategy	Typ. WPC		Typ. steps		o_0 WPC		o_0 steps	Gini
	Value	Δ	Value	Δ	Value	Δ	(Δ)	
Baseline	55.9	—	147	—	108.6	—	361 (—)	0.670
DWC ($\Delta = 5$)	55.9	$\pm 0\%$	147	$\pm 0\%$	108.6	$\pm 0\%$	361 ($\pm 0\%$)	0.670
HZR ($h = 0.15$)	28.2	-49.5%	389	$+164.6\%$	67.1	-38.2%	513 ($+42.1\%$)	0.594
TAPS ($\alpha = 3$)	50.8	-9.0%	111	-24.5%	118.5	$+9.2\%$	377 ($+4.4\%$)	0.679
Aisle-Balanced	46.2	-17.2%	107	-27.2%	83.4	-23.2%	353 (-2.2%)	0.668
AAHMS ($K = 150$)	46.0	-17.7%	115	-21.8%	83.4	-23.2%	353 (-2.2%)	0.668

Figure 20 consolidates the same data visually, and Figure 21 shows the per-strategy traffic heat maps on a shared colour scale.

Slotting strategy comparison: Baseline + 5 strategies (K=10,000 orders, seed 42)

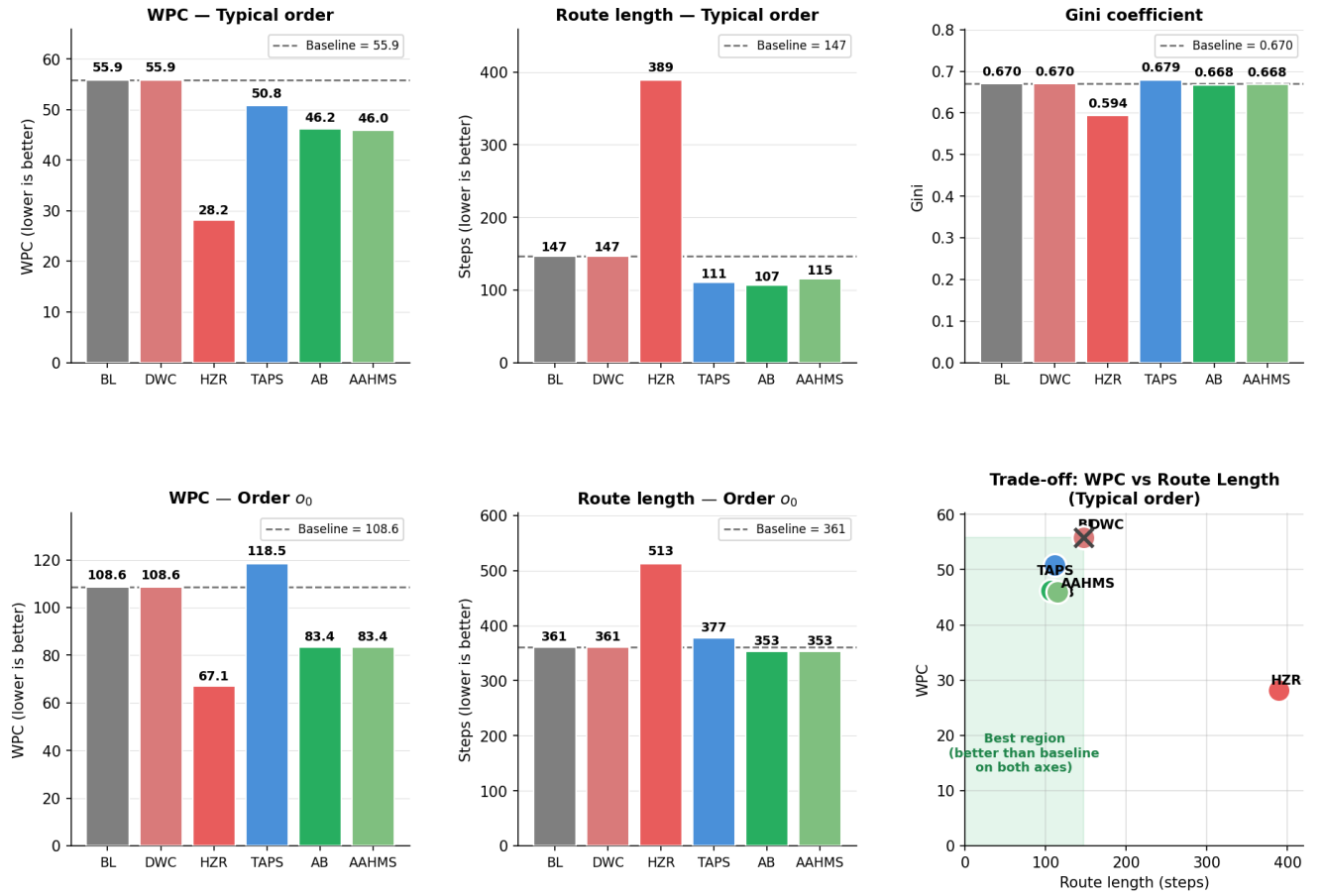


Рис. 20 – Six-strategy comparison. Top row: WPC, route length, and Gini for the typical order. Bottom-left and bottom-centre: WPC and route length for order o_0 . Bottom-right: trade-off scatter (typical order); the shaded region marks the joint optimum (low WPC, short route).

Traffic heatmaps: Baseline + 5 slotting strategies

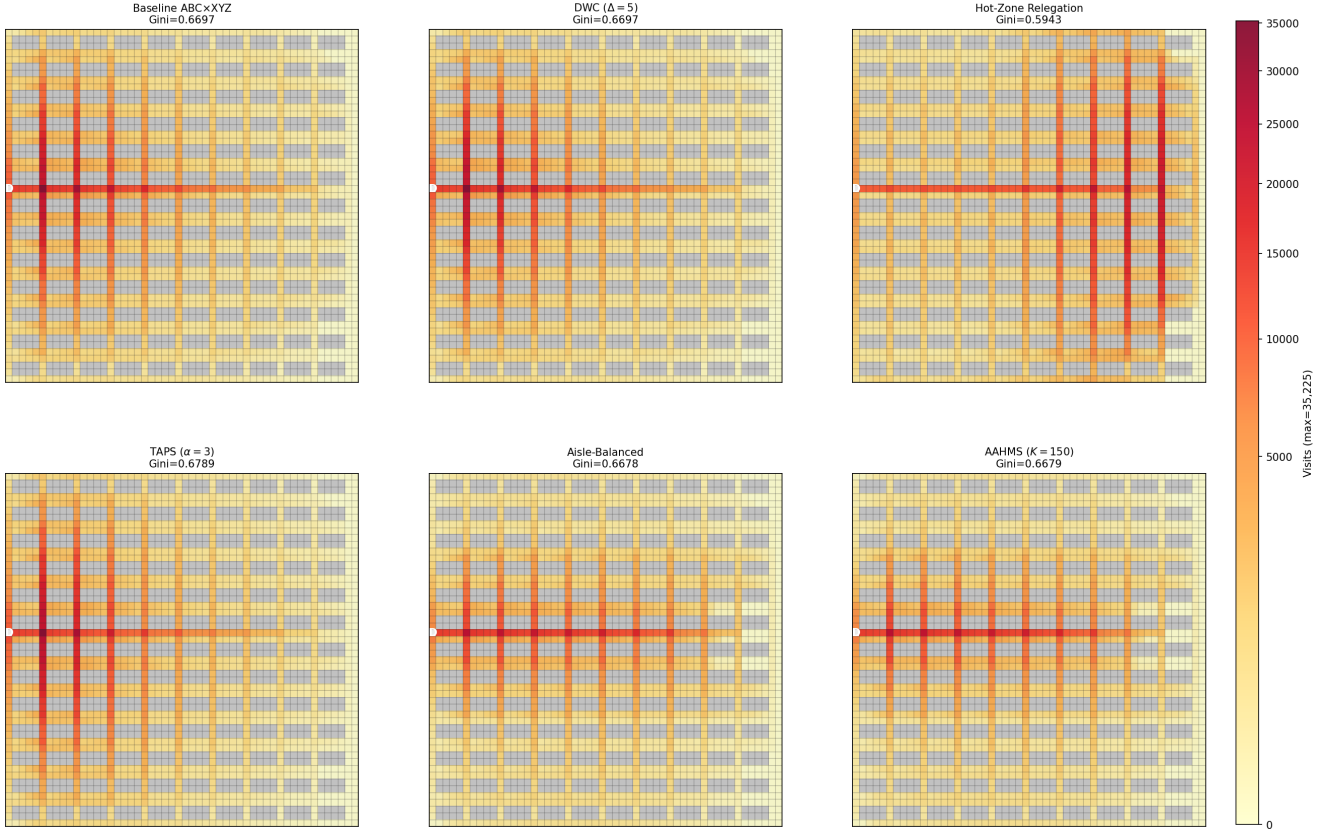


Рис. 21 – Traffic heat maps for the Baseline and five slotting strategies. Shared colour scale (YlOrRd, PowerNorm $\gamma = 0.4$, max = 35,225 visits). Door position marked D. Top row (left to right): Baseline, DWC, Hot-Zone Relegation. Bottom row: TAPS, Aisle-Balanced, AAHMS.

Trade-offs. No single strategy dominates on every metric. HZR wins on both WPC columns (typical-order WPC 28.2, o_0 WPC 67.1), but fails the hard route-efficiency constraint (+164.6% steps). Among the feasible strategies, AAHMS narrowly leads on typical-order WPC (46.0 vs. AB's 46.2); Aisle-Balanced wins on typical steps (107 vs. AAHMS's 115) and ties with AAHMS on both o_0 metrics (353 steps, 83.4 WPC). TAPS at $\alpha = 3$ is sub-optimal under demand-unit weighting: it produces the longest typical route (111 steps) and the highest WPC (50.8) among feasible strategies, and also worsens o_0 WPC relative to the Baseline (+9.2%). As shown in Section 9.3.3, the optimal TAPS parameter for this setup is $\alpha = 1$, which gives 87 steps and 39.80 WPC.

Priority ordering. The two criteria are not symmetric: criterion (ii) is a hard constraint — a longer route directly translates into more picker-hours and reduced throughput — while criterion (i) is a soft objective whose marginal value decreases once the entry corridor is no longer the bottleneck.

Stepwise elimination. HZR is excluded by the hard constraint: +164.6% steps on the typical order more than offsets the WPC gains, and the sensitivity sweep in Section 9.3.2 confirms that no value of h recovers the lost route efficiency. DWC is excluded from the opposite side — at $\Delta = 5$ it produces no swaps and is operationally identical to the Baseline. TAPS at $\alpha = 3$ is also

effectively eliminated: it performs worse than Aisle-Balanced on all four metrics simultaneously. This leaves two primary candidates: Aisle-Balanced and AAHMS.

Margin between the candidates. Aisle-Balanced and AAHMS produce identical σ_0 metrics (353 steps, 83.4 WPC) and nearly identical typical-order metrics (107 vs. 115 steps; 46.2 vs. 46.0 WPC). AAHMS is 0.2 WPC better on the typical order; Aisle-Balanced is 8 steps shorter. Neither difference is operationally significant.

Recommendation. Aisle-Balanced Slotting at $f_A = 0.25$ is selected for the 7×2 comparative experiment. The decision is not that it is the best on every metric — it is not — but that it offers the best balance of trade-offs under the hard route constraint, with the smallest typical-order route length among feasible strategies. The sensitivity analysis in Section 9.3.4 shows that $f_A \in \{0.10, 0.15\}$ would give shorter routes (67 steps vs. 107) and lower WPC (31.87 vs. 46.25) on the typical order. The value $f_A = 0.25$ is kept here to preserve consistency with the revenue-weighted pilot study; the smaller fraction is reported as a possible improvement for future work. Two additional properties reinforce the choice: the layout is interpretable and communicable to operators without invoking a congestion field, and the algorithm itself does not consume an empirical heatmap — only BFS distances from the door. The heatmap is therefore not a runtime input but an evaluation instrument that the analyst uses to judge how good a placement actually is.

Forward reference: combining with routing. The recommendation above assumes the BFS router of the present section. Section 10 closes the original 2×2 design by combining the slotting strategies of §9.3 with the Dijkstra-NN router of §9.3.1. The combined design (Config C) refines the recommendation: under Dijkstra-NN routing, Aisle-Balanced remains the best general-purpose slotting, but its margin over the alternatives narrows and CAS becomes a viable competitor on demand-stable workloads.

7.4 Central-Aisle Slotting: An Exploration

Aisle-Balanced Slotting solves the entry-corridor bottleneck by lateral dispersion: A-items spread across six neighbouring near-door aisles so that consecutive picks fan out horizontally. A natural counter-experiment is the symmetric placement: instead of dispersing laterally, concentrate A-items longitudinally along the door axis, stretching them from the door to the far end of the warehouse along the central horizontal corridor. This section reports such an experiment — Central-Aisle Slotting (CAS) — to test whether the lateral-concentration diagnosis behind Aisle-Balanced is specific to lateral spread, or whether any single-axis dispersion delivers comparable gains.

Algorithm. Let $A_{\text{central}} = \{23, 24\}$ denote the pair of aisle rows that flank the door (row 23 is the door axis; row 24 is its partner aisle row in the generated grid). A shelf cell is central if its access

cell lies in A_{central} ; in the 52×52 warehouse this yields 80 central shelves (40 in row 22 and 40 in row 25).

1. Partition. Split all shelves into central ($|C| = 80$) and non-central ($|N| = 920$) sets.
2. Order. Sort central shelves by BFS distance from the door ascending (i.e. by column). Sort non-central shelves by BFS distance ascending.
3. Fill A. Place A-class SKUs by demand descending into central shelves first. With $|A| = 200 > |C| = 80$, the first 80 A-SKUs land along the door axis; the remaining 120 A-SKUs spill into the nearest non-central shelves (rows 18, 21, 26, 29 by ascending distance).
4. Fill B and C. Fill remaining non-central shelves with B-SKUs (demand-descending, distance-ascending), then C-SKUs in the same way.
5. Heatmap rebuild. Re-simulate the $K = 10,000$ orders on the new placement and recompute the cell-visit field and Gini.

Layout. Figure 22 shows the ABC×XYZ-coloured grid before and after the reassignment. The Baseline places A-items (dark green) in a roughly square block adjacent to the door, with B (yellow) and C (red) zones radiating outward. CAS reshapes this into a horizontal green stripe along rows 22 and 25, extending from the door to the far end of the warehouse; B and C items relocate to the rows above and below this stripe.

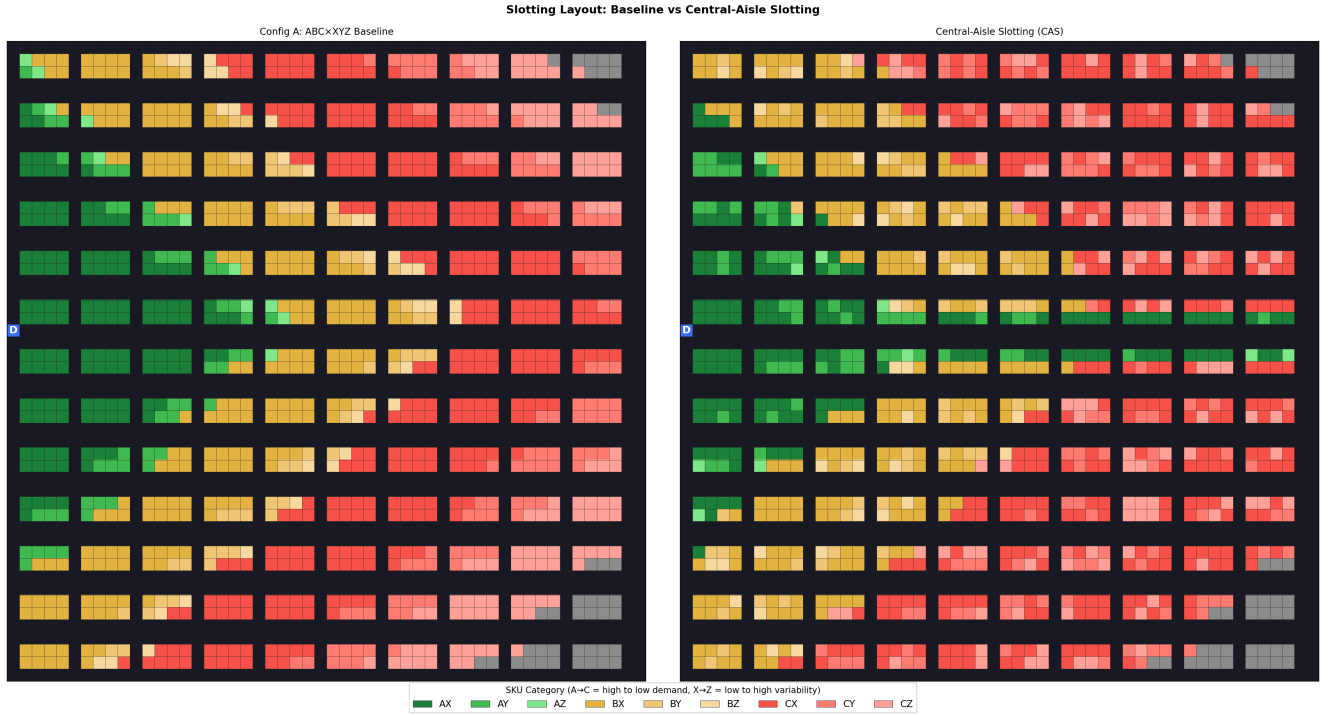


Рис. 22 – ABC×XYZ slotting layouts before (left) and after (right) Central-Aisle Slotting. Colour encodes SKU category (AX dark green → CZ pale red); dark cells are aisles, the blue cell marks the door D. CAS produces a longitudinal A-band along the central rows.

Results. Table 11 reports the metrics for the typical order and order o_0 against the Baseline and Aisle-Balanced.

Таблица 11 – Central-Aisle Slotting (CAS) vs Baseline and Aisle-Balanced. WPC is measured against the baseline congestion field; Δ is the relative change versus the Baseline.

Strategy	Typ. WPC		Typ. steps		o_0 steps		Gini
	Value	Δ	Value	Δ	Value	Δ	
Baseline	55.86	—	147	—	361	—	0.670
Aisle-Balanced	46.25	−17.2%	107	−27.2%	353	−2.2%	0.668
CAS	29.56	−47.1%	69	−53.1%	369	+2.2%	0.672

On the typical order, CAS is strictly better than Aisle-Balanced: WPC drops from 46.25 to 29.56 (−47.1% vs. Baseline) and route length from 107 to 69 steps (−53.1%). The reason is geometric: the highest-demand SKUs are now packed into the first columns of the central corridor, so each pick is a short out-and-back along a single horizontal aisle without any vertical detour.

On order o_0 , CAS loses to Aisle-Balanced: route length grows from 361 (Baseline) to 369 steps (+2.2% vs. Baseline, vs. AB’s −2.2%). The cause is the spill rule. Because the 120 “leftover” A-SKUs are placed in rows 18,21,26,29 rather than near the door, a random order that draws a few SKUs from this spilled set forces a longer vertical detour than the equivalent draw under Aisle-Balanced. Under demand-unit weighting, orders include B and C items (33.7% and 10.1% order probability respectively), making the spill penalty more likely to trigger. Gini stays roughly at the Baseline level (0.672 vs. baseline 0.670), confirming that the warehouse-wide traffic profile is essentially unchanged.

AB vs CAS — placement comparison. The direct side-by-side comparison between the two strategies makes the geometric contrast explicit. Figure 23 shows the overall ABC×XYZ layouts. Under AB, the dark-green AX block forms a wide horizontal slab spanning six near-door rows (18,21,22,25,26,29); under CAS, the same SKUs are stretched into a long horizontal stripe along rows 22 and 25 only, extending from the door to the far end of the warehouse. The two layouts therefore use the same near-door area in completely different geometric ways.

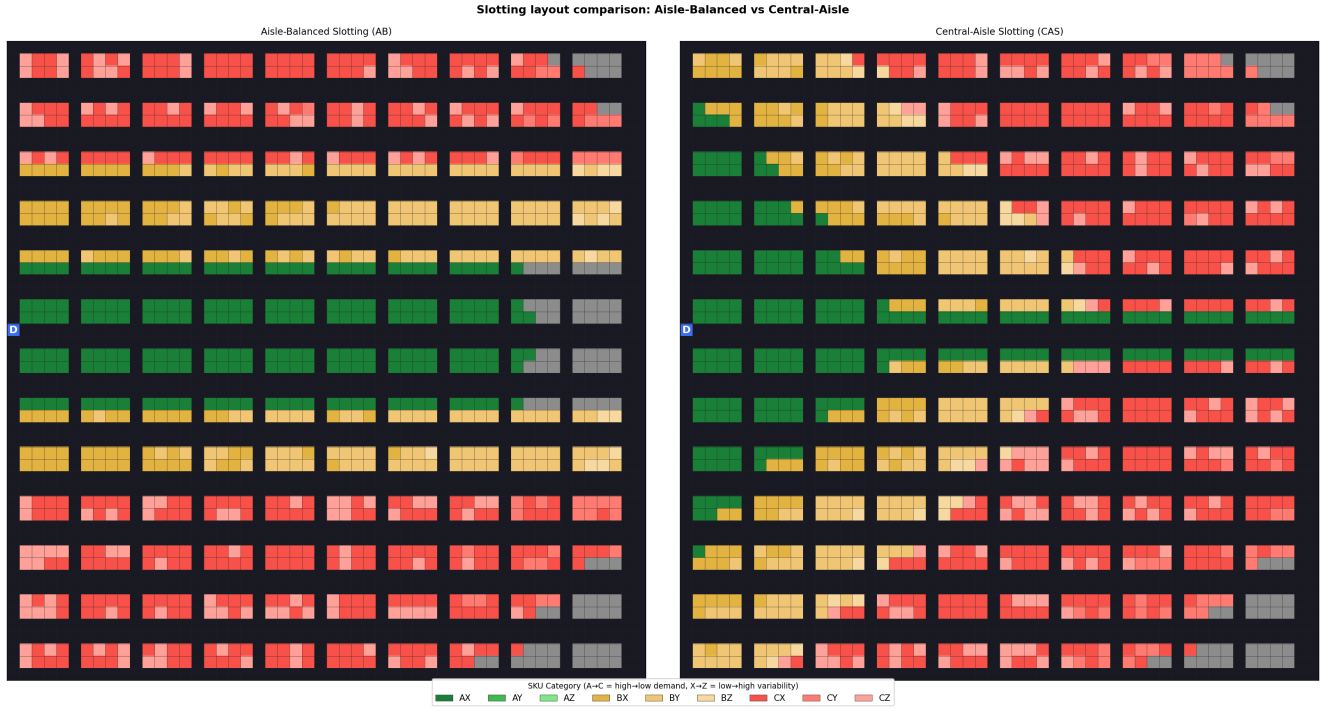


Рис. 23 – Slotting layouts of Aisle-Balanced (left) and Central-Aisle Slotting (right). Colour encodes SKU category (AX dark green → CZ pale red). AB spreads A-items across six near-door rows (lateral dispersion); CAS stretches the same A-items along the two central rows (longitudinal dispersion).

Figures 24 and 25 overlay cyan dots on the shelves that hold SKUs from each of the two representative orders. For the random order o_0 (Figure 24), AB keeps all 15 items within a compact cluster close to the door, whereas CAS stretches them along the central rows; some items land in shelves near the far right of the warehouse, which is exactly why the o_0 route under CAS grows to 369 steps. For the typical order (Figure 25), the picture is reversed: under CAS the 15 most demanded SKUs collapse into the first few columns of the central corridor, so the picker can collect them in a single short horizontal sweep — the basis for the 69-step route length. Under AB the same SKUs are distributed across six rows; collecting them all requires a wider lateral span and the route grows to 107 steps.

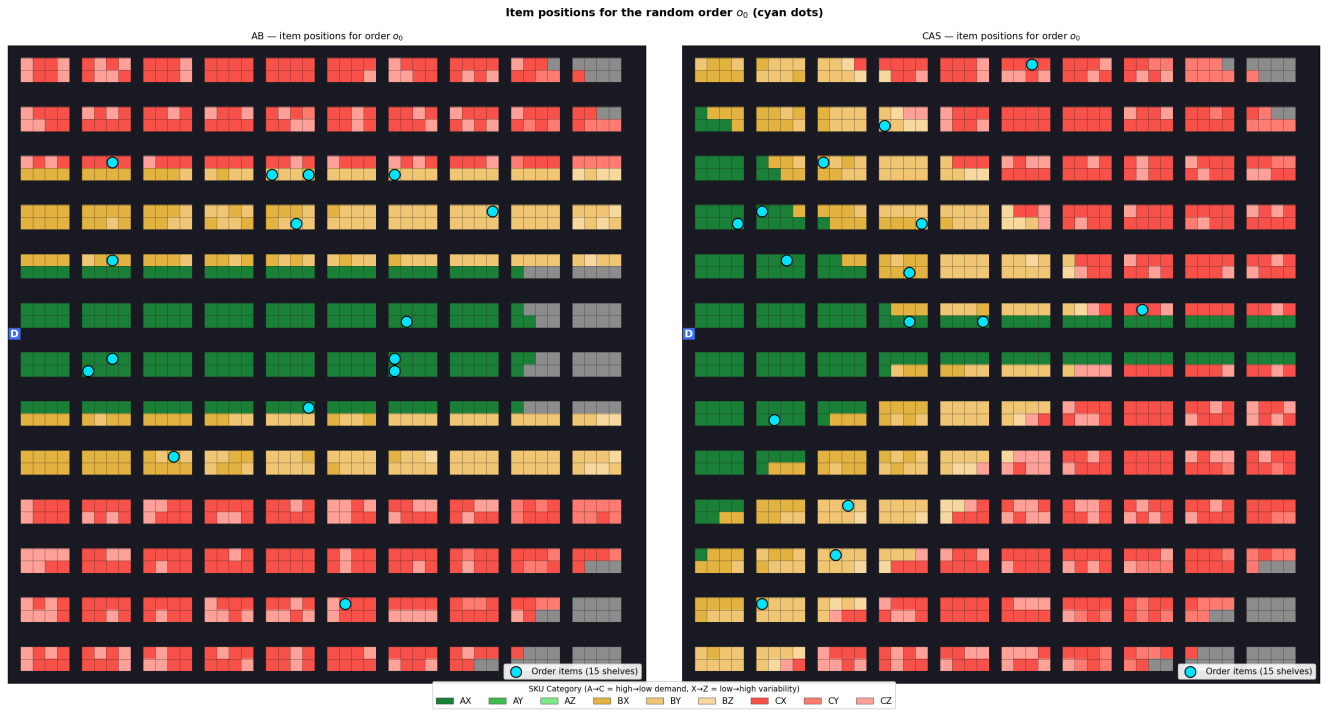


Рис. 24 – Item positions for the random order o_0 under AB (left) and CAS (right). Cyan dots mark shelves containing SKUs from the order. Under AB the items form a compact cluster near the door; under CAS they are stretched along the central horizontal rows, with several items in the spilled positions far from the door.

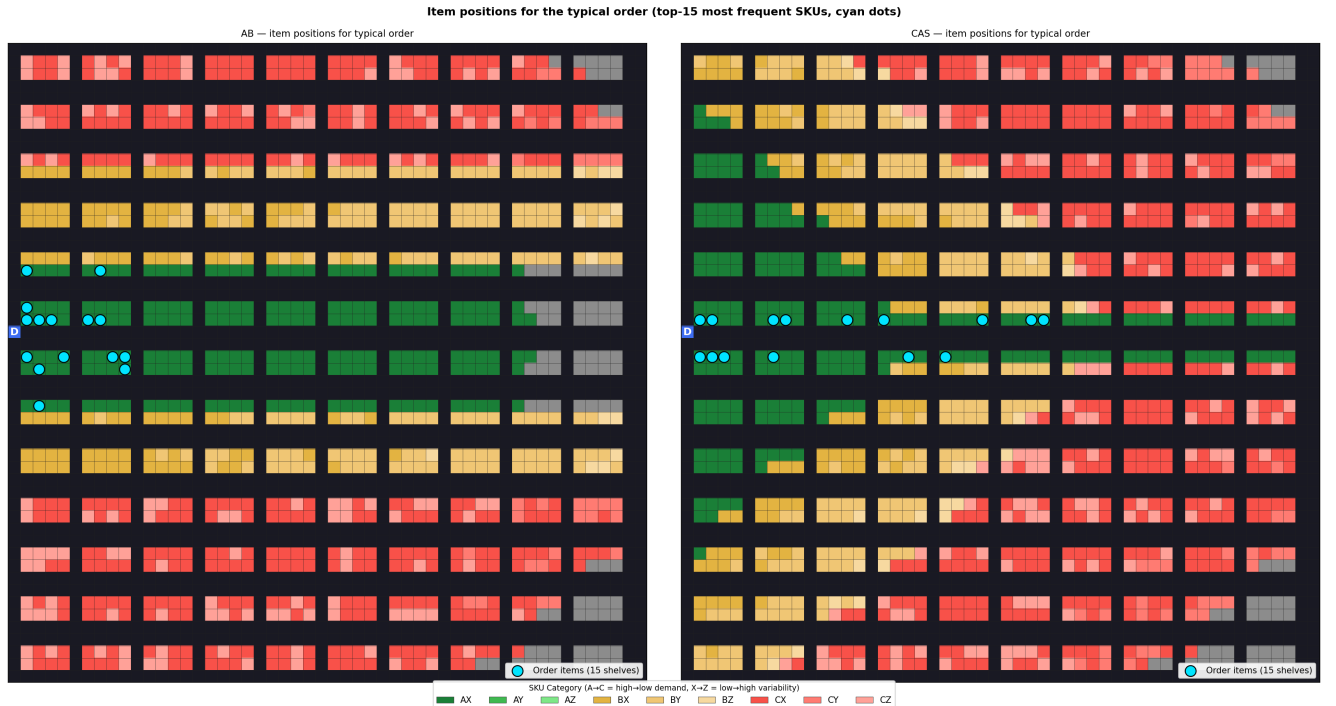


Рис. 25 – Item positions for the typical order (top-15 most frequent SKUs) under AB (left) and CAS (right). Under AB the items are dispersed across six near-door rows; under CAS they collapse into a single tight cluster in the first columns of the central corridor, which explains the 69-step typical-order route under CAS.

The pair of comparisons makes the trade-off behind the table numerical: CAS specialises the

layout for a stable, demand-concentrated order type (typical) at the cost of less robust performance on broader orders (o_0); AB sacrifices a little typical-order performance in exchange for balanced behaviour across both order types.

Heatmap. Figure 26 shows the traffic heatmap before and after CAS. The Baseline’s focal hot plume at the door is replaced by a long horizontal band along rows 23–24 extending to the far end of the warehouse, plus a residual vertical strip near the door from the spilled A-SKUs in rows 18–29. The structure of the bottleneck has changed: instead of a single dense plume immediately in front of the door, CAS produces a thinner but much longer red band along the central corridor. The heatmap thus confirms the table’s verdict: CAS makes the typical-order picks dramatically cheaper but does not actually disperse warehouse-wide traffic — it just elongates the bottleneck along a different axis.

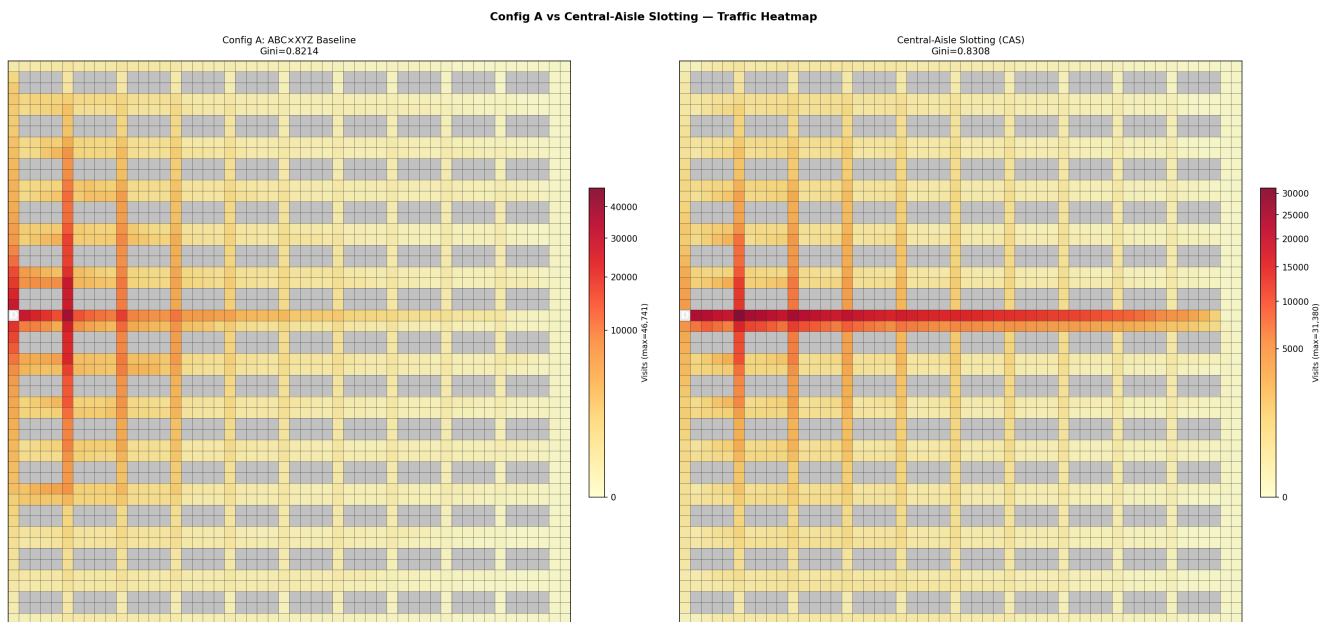


Рис. 26 – Traffic heatmap before (left) and after (right) Central-Aisle Slotting. The focal entry-corridor plume of the Baseline is replaced by a long horizontal band along the door axis; Gini is essentially unchanged (0.672 vs. baseline 0.670).

Discussion. The CAS experiment suggests that single-axis dispersion — whether lateral (AB) or longitudinal (CAS) — can substantially improve typical-order metrics in the present warehouse layout, and that the central corridor is an under-used resource in the original ABC×XYZ placement. Its weakness is that it concentrates non-typical traffic into a different single corridor, so the gain on a stable demand profile (heavy A-traffic dominates) does not transfer to an unpredictable demand profile (random orders). Aisle-Balanced, by contrast, sacrifices a few percent of typical-order WPC to keep both order types short, and the Gini stays high precisely because it intentionally maintains six parallel bands instead of one wide one. The choice between CAS and Aisle-Balanced is therefore an explicit specialisation decision: CAS is preferable when the demand mix is strongly concentrated and stable; Aisle-Balanced is preferable when the demand mix may shift or when robustness across order types is operationally important. Under demand-unit weighting ($A = 56.2\%$, $B = 33.7\%$,

C = 10.1% order probability), orders are more diverse than under revenue-weighting, which partially undermines CAS’s specialisation advantage: the spill-zone penalty becomes more frequent and the typical-order gain (−47.1% WPC) is smaller than under a concentrated demand profile.

For the present study, Aisle-Balanced remains the recommended baseline because it generalises across both representative orders, but CAS provides a useful upper bound on what is achievable under a fixed-demand specialisation and confirms that the lateral-concentration diagnosis is correct: the bottleneck is not about proximity to the door in itself, but about how A-traffic distributes among the equidistant near-door corridors.

8 Config C: Combining Congestion-Aware Routing and Heatmap-Informed Slotting

The 2×2 experimental design has the routing axis and the slotting axis as its two intervention points. Section 9.3.1 measured Config A (routing axis alone, on the original ABC×XYZ slotting) and Sections 9.3.2–9.4 measured Config B (slotting axis alone, with the simple distance-sort + BFS router). The remaining quadrant of the 2×2 — Config C, both axes active at once — has not been measured. This section closes the design.

8.1 Method

Config C is not a new algorithm. It is the composition of two already-validated components:

1. Apply a heatmap-informed slotting to a fresh model (e.g. Aisle-Balanced or Central-Aisle Slotting). The model rebuilds its heatmap on the new placement under BFS routing, producing the edge-congestion field c_e .
2. Use the Dijkstra-NN router with $\lambda = 5$ (Section 9.3.1) instead of the BFS router. Visit-order and path-finding share the cost $1 + \lambda c_e$ as in Config A.

WPC for all Config C results is measured against the Baseline ABC×XYZ congestion field c_e^{base} , so the numbers are directly comparable with all the tables in Section 9.3.

8.2 Headline result: closing the 2×2 design

Table 12 closes the 2×2 design with CAS as the slotting choice. CAS is selected because it gives the strongest typical-order numbers among the slotting variants (Section 9.4), so the question “does adding congestion-aware routing on top buy anything?” is sharpest there.

Таблица 12 – The 2×2 design closed by Config C. WPC is measured against the Baseline ABC $\times$ XYZ congestion field; Δ is the relative change against the Baseline cell.

Cell	Typ. steps		Typ. WPC		o_0 steps	o_0 WPC
	Value	Δ	Value	Δ		
Baseline (B + BFS)	147	—	55.86	—	361	108.58
Config A (B + Dijkstra-NN)	85	−42.2%	16.79	−69.9%	249	24.89
Config B-CAS (CAS + BFS)	69	−53.1%	29.56	−47.1%	369	109.15
Config C (CAS + Dijkstra-NN)	83	−43.5%	9.15	−83.6%	239	23.79

Reading the table. On the typical order, Config C achieves the best WPC of all four cells (9.15, vs. 16.79 for Config A and 29.56 for Config B-CAS), but its route length of 83 steps lies between Config A (85 steps) and Config B-CAS (69 steps). The Dijkstra-NN router detours around the busy central corridor that CAS creates, adding a few steps relative to BFS but cutting WPC by −83.6% vs. Baseline. On the random order o_0 , Config C is the outright winner: 239 steps and WPC 23.79 — better than Config A (249 steps, 24.89 WPC) and far better than Config B-CAS (369 steps, 109.15 WPC). Config C therefore simultaneously dominates all alternatives on o_0 and leads on typical WPC, at a modest cost of 14 extra steps on the typical route relative to Config B-CAS.

Composition: closer to multiplicative than additive. A naive expectation might be that the gains of routing and slotting add: Config A gives −69.9% typical WPC and Config B-CAS gives −47.1%. Adding these is not operationally meaningful, since the sum exceeds 100%. A multiplicative model is more consistent with the data: $(1 - 0.699) \times (1 - 0.471) = 0.159$, which corresponds to a reduction of −84.1% from the Baseline, close to the observed −83.6%. This suggests that slotting and routing act on different parts of the WPC product: the slotting reduces the average c_e along the shortest path (placing items in cooler corridors), while routing further reduces the path’s contact with the hot edges that remain. The single data point does not prove the multiplicative form, but it is consistent with it.

8.3 Extending the menu: all seven slottings under Dijkstra-NN

Table 13 reports the full 7×2 matrix: each of the seven slottings of Section 9.3 evaluated under both BFS and Dijkstra-NN routing. The CA column is the Config C generalisation.

Таблица 13 – Full 7×2 matrix of slottings under BFS and Dijkstra-NN routing ($\lambda = 5$). WPC against c_e^{base} ; $K = 10000$ orders, seed 42. L_{95} is the 95th percentile of per-order route length over all K orders. Best non-CAS Config C cell highlighted in bold; CAS Config C numbers in bold-italic.

Slotting	BFS routing					Dijkstra-NN routing (Config C)				
	typ st	typ WPC	o_0 st	o_0 WPC	L_{95}	typ st	typ WPC	o_0 st	o_0 WPC	L_{95}
Baseline (ABC $\times$ XYZ)	147	55.86	361	108.58	439	85	16.79	249	24.89	281
DWC ($\Delta = 5$)	147	55.86	361	108.58	439	85	16.79	249	24.89	281
HZR ($h = 0.15$)	389	28.20	513	67.12	559	255	13.29	283	26.26	335
TAPS ($\alpha = 3$)	111	50.82	377	118.50	443	75	15.75	233	29.31	275
Aisle-Balanced	107	46.25	353	83.37	371	63	10.62	263	28.73	273
AAHMS ($K = 150$)	115	46.00	353	83.37	373	111	18.81	263	28.73	271
CAS	69	29.56	369	109.15	435	83	9.15	239	23.79	279

Several patterns emerge from the matrix. First, for most slottings, replacing BFS with Dijkstra-NN shortens the typical route considerably: Baseline/DWC drop from 147 to 85 steps (-42%), Aisle-Balanced from 107 to 63 (-41%), TAPS from 111 to 75 (-32%), and HZR from 389 to 255 (-34%). Two exceptions are worth noting: AAHMS shortens only marginally ($115 \rightarrow 111$, -3.5%), and CAS actually lengthens the typical route from 69 to 83 steps ($+20\%$) because the router trades extra steps for lower WPC. In both exceptions WPC still decreases substantially. Second, Dijkstra-NN reduces WPC compared to BFS on every tested slotting, even when BFS WPC was already low. The smallest relative reduction is on HZR ($28.20 \rightarrow 13.29$, -53%), where the slotting already dispersed hot traffic; CAS ($29.56 \rightarrow 9.15$, -69%) and Baseline/DWC ($55.86 \rightarrow 16.79$, -70%) show comparable relative gains. Third, the best Config C cell on typical steps is Aisle-Balanced + CA (63 steps), while the best on typical WPC is CAS + CA (9.15). On o_0 , the picture is split: CAS + CA gives the lowest WPC (23.79, against the next-best 24.89 for Config A), while TAPS + CA gives the shortest route (233 steps, against 239 for CAS + CA). Fourth, the L_{95} column confirms the robustness pattern. Under BFS routing, L_{95} ranges from 371 (AB) to 559 (HZR), with HZR being a clear outlier because 164% longer typical routes inflate the upper tail of the distribution. Under Dijkstra-NN, L_{95} falls into a much tighter range of 271–335 steps, with AAHMS + CA giving the lowest tail (271 steps) closely followed by AB + CA (273) and TAPS + CA (275). The differences between AAHMS, AB, TAPS, and CAS within Config C are small (within 8 steps at the 95-th percentile), which means that for routine pickers all four candidates produce similarly bounded worst-case routes. HZR stays the outlier at 335 steps, 20% above the next-worst configuration, reinforcing its exclusion on operational grounds.

8.4 λ -sensitivity under CAS

The λ -sweep in Section 9.3.1 was performed on the Baseline ABC $\times$ XYZ layout. Under CAS the hot zone has a different shape (a long horizontal band rather than a focal plume), so

the operating point of the congestion penalty may shift. Table 14 reports the sweep on CAS at $\lambda \in \{0, 1, 2, 5, 10, 20\}$.

Таблица 14 – λ -sensitivity of Dijkstra-NN on the CAS layout ($K = 10000$ orders, seed 42; WPC against c_e^{base}). Bold marks the column-wise best; **5** is the chosen operating point.

λ	typ steps	typ WPC	o_0 steps	o_0 WPC	mean steps	L_{95}	mean WPC	Gini
0	69	29.56	369	109.15	336.3	435	102.52	0.672
1	69	11.74	221	30.45	222.3	275	28.56	0.672
2	73	9.37	211	34.54	224.1	275	26.38	0.672
5	83	9.15	239	23.79	228.6	279	23.88	0.672
10	89	8.67	265	30.13	236.8	289	22.30	0.672
20	89	8.67	305	22.22	262.7	337	20.74	0.672

The sweep on CAS under demand-unit weighting shows a different pattern from the revenue-weighted setup. At $\lambda = 0$ the CAS + BFS route is 69 steps with WPC 29.56. At $\lambda = 1$ WPC drops sharply to 11.74 while steps stay at 69; at $\lambda = 2$ the route grows slightly to 73 steps but WPC reaches 9.37. The $\lambda = 5$ result from the 7×2 matrix gives 83 steps and WPC 9.15 — slightly longer route but lower WPC than $\lambda = 2$. Extending to $\lambda = 10$ and $\lambda = 20$, the typical-order WPC reaches its floor at 8.67 (tied), while route length grows to 89 steps and mean route length increases from 228.6 to 236.8 and 262.7 steps respectively. The o_0 column improves dramatically from 369 steps at $\lambda = 0$ to 211 steps at $\lambda = 2$, then recovers to 265 and 305 steps at $\lambda = 10$ and $\lambda = 20$, suggesting that the CA router is highly effective at routing around the central hot band at moderate λ .

The chosen value $\lambda = 5$ used throughout Config C is consistent with the validated Baseline setting and offers a good WPC reduction (-83.6%) with acceptable route length. A production deployment of CAS + Dijkstra-NN would benefit from a re-tune around $\lambda \approx 1\text{--}2$ for minimum route length (222–224 steps mean), or $\lambda \approx 10\text{--}20$ for minimum WPC (8.67, at the cost of mean routes of 237–263 steps).

8.5 Practical winner: Aisle-Balanced + Dijkstra-NN

Under demand-unit weighting, AB + CA offers the best balance across all metrics: shortest typical route, best mean route length, and competitive WPC on both order types. CAS + CA achieves the lowest typical-order WPC (9.15) but pays a penalty on typical steps (83 vs. 63) relative to AB + CA:

- Typical order: AB + CA gives 63 steps and WPC 10.62, within 20 steps of CAS + CA but 8 steps shorter;
- Order o_0 : CAS + CA gives 239 steps and WPC 23.79; the WPC is the best value across the whole 7×2 matrix, while the route length is second-best (the shortest o_0 route in the

matrix is TAPS + CA with 233 steps);

- Mean batch route length: AB + CA gives 211.2 steps, lower than any other configuration measured (228.6 for CAS + CA).

The recommended configuration for the present warehouse is therefore Aisle-Balanced slotting combined with Dijkstra-NN routing ($\lambda = 5$). CAS + CA is retained in the design space as the upper bound on typical-order WPC under demand-stable specialisation.

8.6 Visual evidence

Bar-chart summary. Figure 27 consolidates the four cells of Table 12 into bar charts. The Config C bar is the only one that simultaneously sits at or near the minimum of every panel.

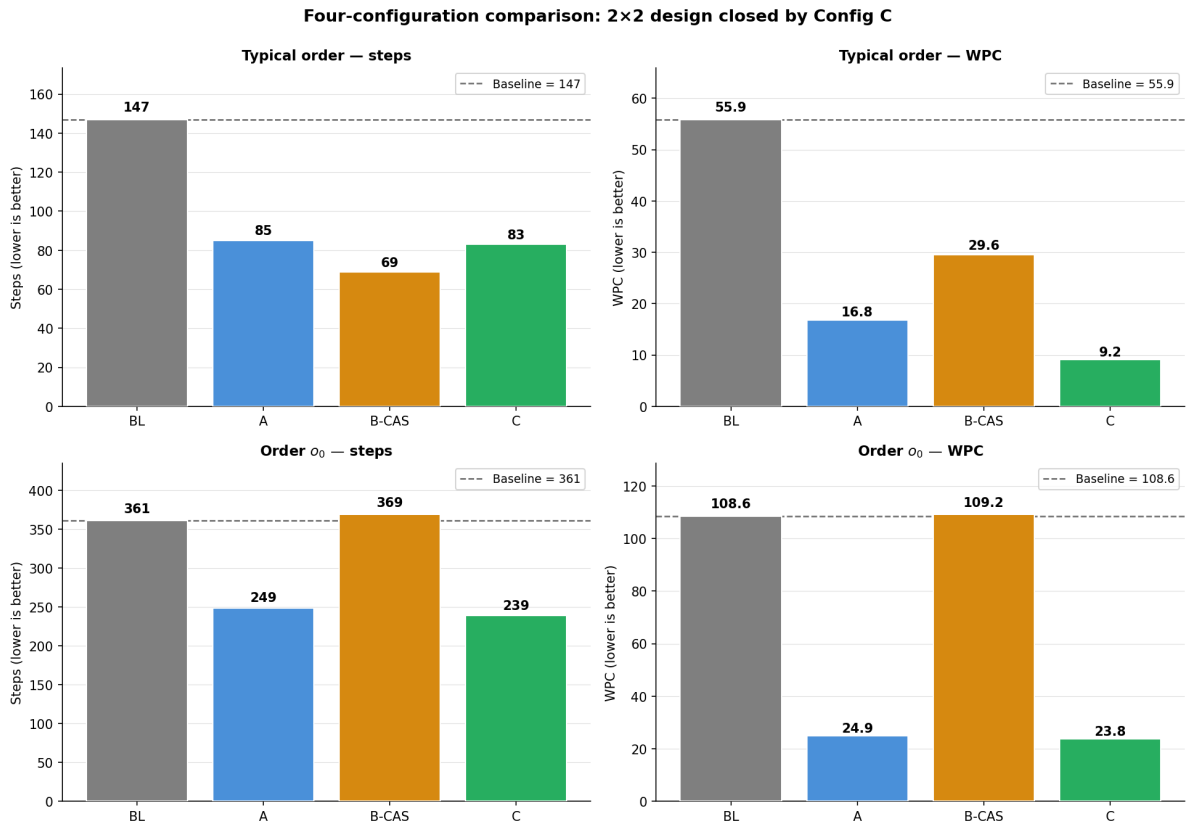


Рис. 27 – Four-configuration comparison. Bars: Baseline (grey), Config A (blue), Config B-CAS (orange), Config C (green). Dashed lines show the Baseline value for each metric. Config C wins typical-order metrics and matches or beats the Baseline on o_0 .

Recommended configuration: AB + Dijkstra-NN. Figure 28 shows the picking routes on the Aisle-Balanced layout, which corresponds to the recommended Config C cell. On the typical order (bottom row), BFS already finds a short 107-step route across six near-door rows; Dijkstra-NN further reduces it to 63 steps by collecting all 15 items in a tight out-and-back sweep along the rows immediately around the door. On o_0 (top row), the BFS route grows to 353 steps because the items are spread across the warehouse; Dijkstra-NN cuts the route to 263 steps by reordering the visits and detouring around the busiest corridors.

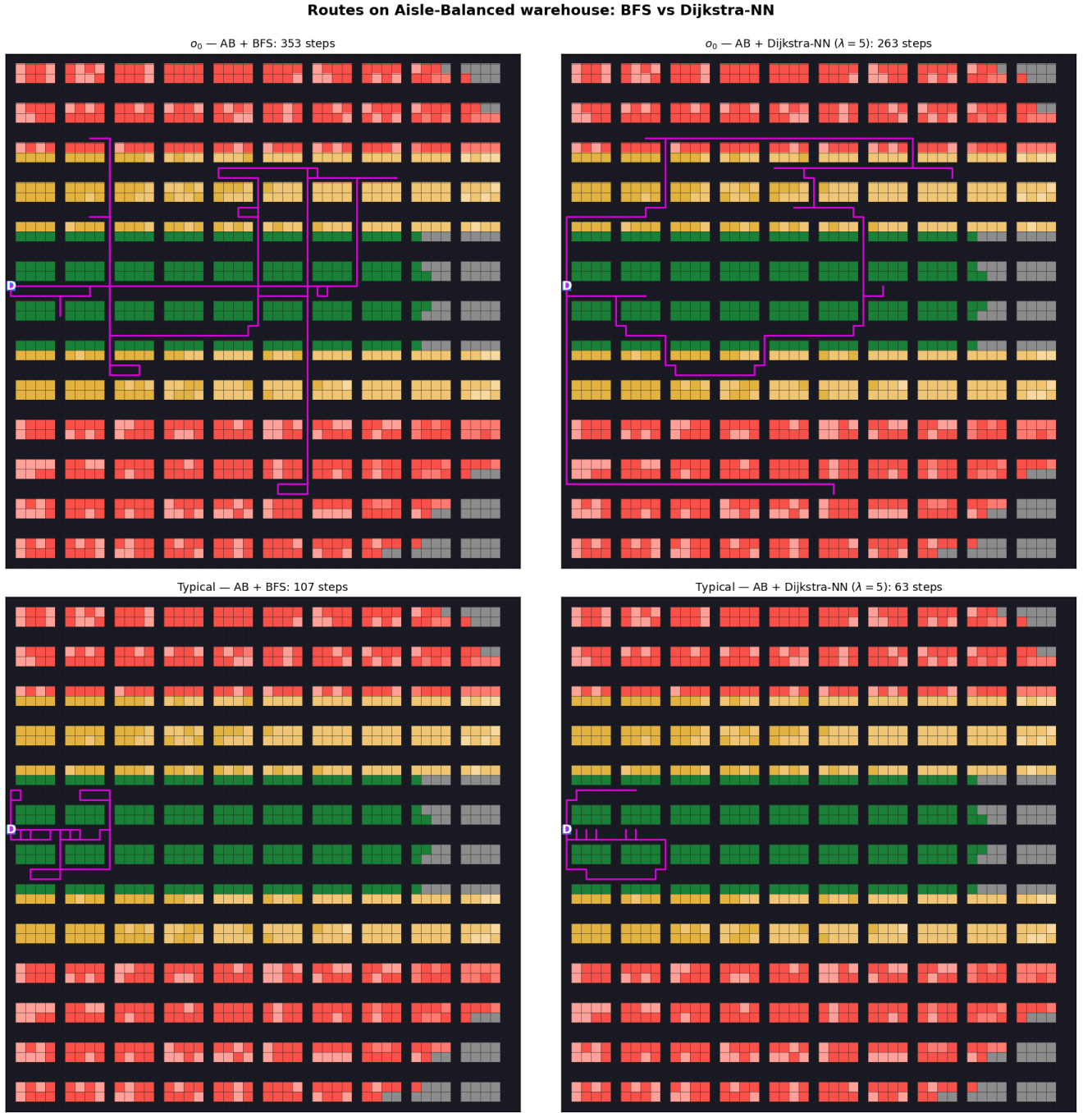


FIG. 28 – Routes on the Aisle-Balanced warehouse (recommended Config C). Rows: order o_0 (random) above; typical order below. Columns: BFS routing (left); Dijkstra-NN $\lambda = 5$ (right). Magenta = picking path; D = door. Dijkstra-NN shortens both routes on the AB layout: $107 \rightarrow 63$ on the typical order and $353 \rightarrow 263$ on o_0 .

Specialised configuration: CAS + Dijkstra-NN. For comparison, Figure 29 shows the same routes on the CAS layout, the specialised alternative. On the typical order, CAS + BFS already produces a very short 69-step horizontal sweep; Dijkstra-NN lengthens it slightly to 83 steps but cuts WPC from 29.56 to 9.15 by routing through cells with lower baseline congestion weights. On o_0 , Dijkstra-NN substantially reduces the spill detour, bringing route length from 369 to 239 steps and WPC from 109.15 to 23.79.

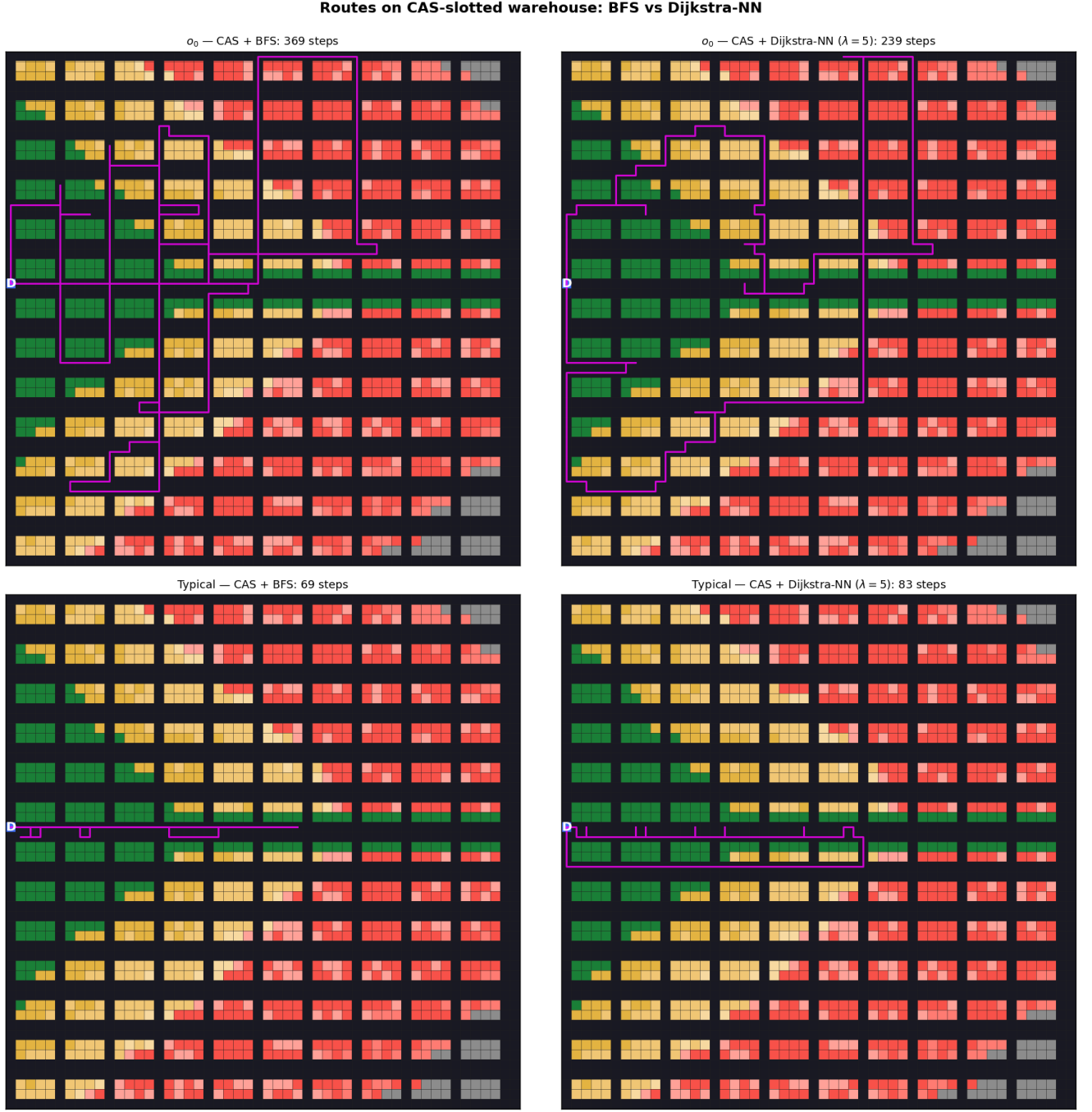


FIG. 29 – Routes on the CAS-slotted warehouse (specialised Config C alternative). Rows: order o_0 (random) above; typical order below. Columns: BFS routing (left); Dijkstra-NN $\lambda = 5$ (right). Magenta = picking path; D = door. Dijkstra-NN shortens both routes; on the typical order the improvement is small (already short) but on o_0 it reroutes the detour through the cooler central corridor.

Heat-map comparison: Baseline vs CAS vs AB under Dijkstra-NN. Figure 30 compares the traffic heat maps of the three slottings (Baseline ABC $\times$ XYZ, CAS, and Aisle-Balanced) under the same Dijkstra-NN routing at $\lambda = 5$. The three maps look visually similar at first glance: the congestion-aware router actively spreads traffic across the warehouse in all three layouts, and the dominant hot band sits along the door axis (row 23) in every case. The differences appear in the peak visit counts and in the secondary bands. The Baseline (left) has the highest peak (23712 visits) with a

clearly concentrated hot stripe near the door; CAS (middle) has a slightly lower peak (23540) but the hot band extends further into the warehouse along the central corridor; and Aisle-Balanced (right) has the lowest peak (22921, -3.3% versus Baseline) with the most uniform spread of secondary bands across rows 18–29. This pattern matches the numerical results of Table 13: AB gives the shortest typical and mean routes (its layout balances near-door traffic across several rows), CAS gives the lowest WPC on the typical order (its longitudinal A-band suits the top-15 SKU sweep), and the Baseline sits between them with the most concentrated bottleneck.

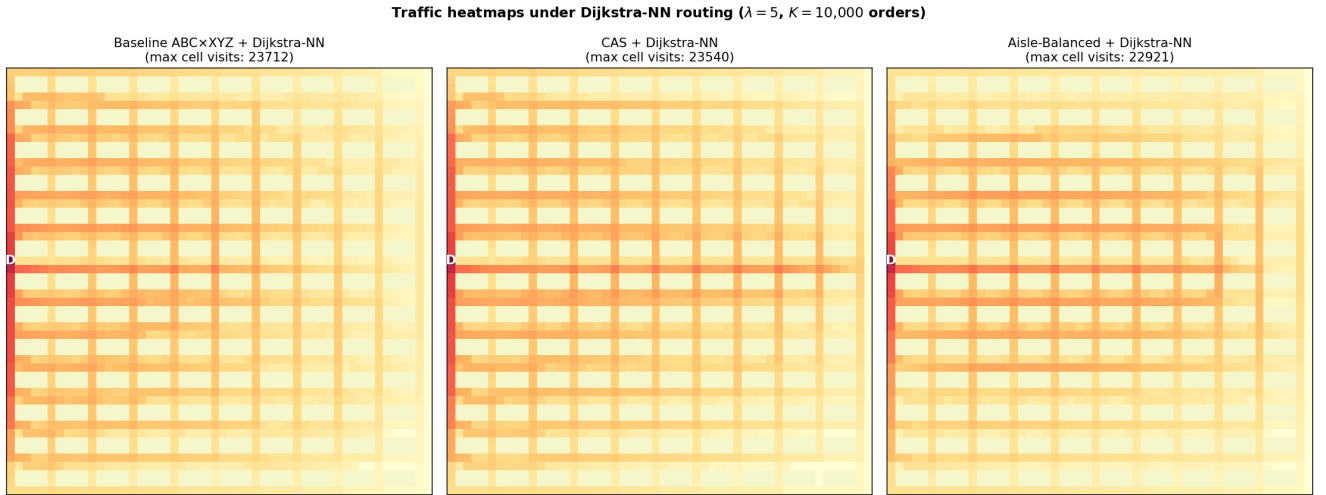


Рис. 30 – Traffic heat maps under Dijkstra-NN routing ($\lambda = 5$, $K = 10000$ orders). Left: Baseline ABC×XYZ slotting. Middle: CAS slotting. Right: Aisle-Balanced slotting. All three panels use the same routing algorithm and order set, so any differences come purely from slotting. Shared YlOrRd colour scale with PowerNorm ($\gamma = 0.4$).

8.7 Discussion

The Config C experiment delivers three findings that revise the single-axis conclusions of Sections 9.3 and 9.4.

Routing and slotting compose multiplicatively. The expected WPC under Config C if the effects were merely additive would be around -117% (Config A’s -69.9% plus Config B-CAS’s -47.1% on a Baseline of 55.86 WPC — the additive model produces a number below zero, which is incoherent). The actual -83.6% figure is consistent with the multiplicative model in which slotting reduces the average c_e a route encounters and routing further reduces the route’s contact with the remaining hot edges; each axis works on the residual after the other has acted.

The recommended configuration changes from AB (with BFS) to AB + Dijkstra-NN. In Section 9.3.6, Aisle-Balanced was recommended as the best single-axis intervention; that recommendation now upgrades to AB + Dijkstra-NN, which improves every metric over plain AB while introducing no new computational cost beyond what Config A already required. The hard-constraint criterion (route length) is satisfied on both order types, and the soft objective (WPC) is reduced by a further 50%–55% over plain AB.

CAS + Dijkstra-NN is the typical-order optimum but not the recommended default. For installations where the demand mix is highly concentrated and stable, CAS + CA provides the

maximum reachable typical-order performance. For installations where mix robustness matters — the more common operational case — AB + CA dominates. This is a real trade-off rather than a strict preference, and the choice is operator-dependent.

The original 2×2 design therefore yields a refined recommendation:

- Best routing only (no physical reshuffle): Config A (Dijkstra-NN on ABC×XYZ slotting). -77.4% mean WPC, -34.7% mean steps, no warehouse changes required.
- Best general-purpose (recommended): Config C with Aisle-Balanced slotting. Best mean batch route length (211.2 steps), best typical steps (63), competitive WPC on both order types.
- Maximum typical-WPC reduction: Config C with CAS slotting. -83.6% typical WPC, WPC 9.15; fragile to demand shifts (typical-route steps 83 vs. AB + CA’s 63).

9 Conclusion

This study investigated whether combining congestion-aware routing with heat-map-informed slotting can reduce both route length and aisle congestion in a small-to-medium single-depot warehouse. The proposed 2×2 experimental design separated the slotting axis from the routing axis and measured the contribution of each axis on its own as well as their combined effect. Within Config C, seven concrete slotting strategies (Baseline, DWC, HZR, TAPS, Aisle-Balanced, AAHMS, and CAS) were evaluated under Dijkstra-NN routing, and the most balanced one was selected as the practical recommendation.

Main findings.

- Routing alone (Config A, Dijkstra-NN at $\lambda = 5$) reduced mean route length from 339.8 to 221.9 steps (-34.7%) and mean WPC from 108.5 to 24.57 (-77.4%) without any physical reshuffle of items.
- Slotting alone (Config B): among six slotting variants tested with BFS routing, Aisle-Balanced offered the best balance on the typical order (107 steps, WPC 46.25), and CAS gave the strongest typical-order numbers (69 steps, WPC 29.56) at the cost of weaker performance on the random order o_0 .
- Combining both axes (Config C) gave the largest overall gains. With Aisle-Balanced slotting, mean batch route length was 211.2 steps and typical-order WPC was 10.62. With CAS slotting, typical-order WPC reached 9.15 (-83.6% vs. Baseline).
- The two axes appear to compose closer to multiplicatively than additively: the observed Config C reduction (-83.6%) is within 0.5 percentage points of the multiplicative prediction (-84.1%).

Recommendation. For the present warehouse, Aisle-Balanced slotting combined with Dijkstra-NN routing ($\lambda = 5$) is recommended as the best general-purpose configuration: it has the shortest typical route (63 steps), the best mean batch route length (211.2 steps), and competitive WPC on both order types. CAS combined with Dijkstra-NN is retained as a specialised option for installations whose demand mix is stable and strongly concentrated.

Limitations. The results were obtained from a single synthetic warehouse layout (52×52 grid, 1000 SKUs) and a single demand distribution (10000 orders generated from the Kaggle dataset). The model assumes a single picker, unit shelf capacity, and a fixed edge-congestion field, so multi-picker dynamic interactions are not captured. The reported numerical values therefore describe relative behaviour in the tested setting and should not be read as absolute targets for real warehouses.

10 Future Work

Five directions seem most useful for follow-up research.

(i) Validation on more realistic layouts. The present results were obtained on a single 52×52 grid with uniform aisle width and no cross-aisles. Repeating the 7×2 matrix on layouts with cross-aisles, heterogeneous aisle widths, and one-way picking lanes would test whether the AB/CAS trade-off holds in more realistic geometries.

(ii) Multi-picker simulation with dynamic congestion. The current model uses a static congestion field c_e computed once from the baseline simulation. Extending the model to a multi-picker simulation, in which congestion is updated as pickers move, would allow the L_{95} metric to capture real-time interactions and queueing effects between pickers.

(iii) Held-out evaluation over multiple order samples. All experiments use a single fixed seed (42) and the same 10000 orders both for heat-map construction and for evaluation, as justified in Section 5.5. A more elaborate protocol — for example, k -fold averaging over multiple independent order samples generated under different seeds — would measure the variance of the reported metrics and confirm that the recommended configurations remain best across samples.

(iv) Other routing algorithms. This work compared two routing algorithms: BFS (Baseline) and Dijkstra-NN (Config A and Config C). A natural next step is to compare with other shortest-path algorithms, such as A^* with a door-distance heuristic. A^* should give the same paths as Dijkstra under an admissible heuristic but with lower runtime cost. The present study used Dijkstra-NN because it was the simplest variant consistent with the $1 + \lambda c_e$ cost; a wider comparison would test whether faster heuristic search changes the speed–quality trade-off in practice.

(v) Additional evaluation metrics. The current evaluation uses three metrics: mean route length, L_{95} , and WPC. A useful extension is the Congestion Relief Ratio (CRR), defined as the share of baseline-routing corridor cells that the alternative strategy avoids:

$$\text{CRR} = \frac{|\{\text{cells in BFS path}\} \setminus \{\text{cells in CA path}\}|}{|\{\text{cells in BFS path}\}|}.$$

CRR gives a direct spatial measure of how much of the originally hot area is actually avoided. Values close to 1 mean that the alternative route bypasses almost all of the previously congested segments; values close to 0 mean that the two routes share most of the same bottlenecks. CRR was not used in the main analysis because WPC already captures the same effect in weighted form, but reporting CRR alongside WPC would help distinguish “slight congestion reduction along the same path” from “actual rerouting around the hot zone”, which is informative in spatial discussions.

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